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Tianyu Smart POS
Terminals Interface Document

Wuhan Tianyu Information Industry Co., Ltd

October 2021

Revision history

No.	Date	Revision content	Version	Preparation	Review	Approval
01	2017.10.16	Draft	V0.1	Gao Song	Peng Tao	
02	2017.10.24	Modify some interfaces and parameters according to the code	V0.2	Zeng Yahui	Peng Tao	
03	2017.11.01	1、setEMVConfig and setEMVConfig interfaces are shielded 2. Set printing parameters and increase font settings	V0.3	Zeng Yahui	Peng Tao	
04	2017.12.06	Add openPSAMCard, closePSAMCard interface and onOpenPSAMCard, onClosePSAMCard callback	V0.4	Zeng Yahui	Peng Tao	
05	2017.12.07	Add transmitIC, transmitRF interface and onTransmitIC, onTransmitRF callback	V0.5	Zeng Yahui	Peng Tao	
06	2017.12.12	Delete transmit interface, add transmitPSAM interface	V0.6	Zeng Yahui	Peng Tao	
07	2018.07.27	Add interfaces such as reading and writing custom data, transparent transmission, setting key index, data encryption and decryption, password keyboard input result, and corresponding callback	V0.7	Zeng Yahui	Peng Tao	
08	2019.03.14	Added error code Description	V0.8	Guo Chen	Peng Tao	
09	2019.03.15	Added printing script Description	V0.9	Guo Chen	Peng Tao	
10	2020.04.26	New module function flow chart, sequence diagram and uml diagram, script printing example, card reading process module, Appendix II printing	V1.0	Gao Song	Peng Tao	

		instructions				
11	2021.03.12	Added a set of interface for printing elements, which can support multi-national font	V1.1	Zeng Yahui	Peng Tao	
12	2021.03.18	Add Mifare card related interface	V1.2	Zeng Yahui	Peng Tao	
13	2021.09.24	Complete the document content according to the SDK code	V1.3	Zeng Yahui	Peng Tao	
14	2021.10.10	Modify onTransResult parameters	V1.4	Zeng Yahui	Peng Tao	
15	2021.01.05	Add KeyInfo description, add PinPadConfig, ScannerConfig description, add print example, modify firmware upgrade callback	V1.5	Zeng Yahui	Peng Tao	
16	2021.01.27	Optimize document format and add description content	V1.6	Zeng Yahui	Peng Tao	
17	2021.02.11	Optimize document format; PinResult Change to PinPadResult, onGetPinResult Change to onGetPinPadResult, release Change to releaseBeeper	V1.7	Zeng Yahui	Peng Tao	
18	2022.08.04	Add the pinpad interface with index	V1.8	Zeng Yahui	Peng Tao	
19	2022.11.03	Added update plaintext working key interface	V1.9	Zeng Yahui	Peng Tao	
20	2022.11.16	Added getAtr interface	V2.0	Zeng Yahui	Peng Tao	
21	2022.12.02	Add Tag Module	V2.1	Zeng Yahui	Peng Tao	
22	2022.12.08	Add interfaces: startLogcat、stopLogcat、lockDevice、unlockDevice、clearAppCache Delete Interface: getLogcat	V2.2	Zeng Yahui	Peng Tao	
23	2022.12.27	Add interface: getDeviceInformation	V2.3	Zeng Yahui	Peng Tao	

		Delete interfaces: getDeviceVersion、getDeviceInfo				
24	2023.01.12	Add interface: settingsConfiguration	V2.4	Zeng Yahui	Peng Tao	
25	2023.02.03	Add interface: setLanguage	V2.5	Zeng Yahui	Peng Tao	
26	2023.03.24	Delete setPinBlockFormat	V2.6	Zeng Yahui	Peng Tao	
27	2023.10.23	Delete unused interfaces	V2.7	Zeng Yahui	Peng Tao	
28	2023.10.30	Delete unused interfaces, add printText、printBitmap、printBarcode	V2.8	Zeng Yahui	Peng Tao	

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Contents

1. Purpose 14
2. Framework 14
3. Integration instructions 14
   3.1. Development environment 14
   3.2. SDK files 14
   3.3. Import SDK 15
4. SDK description 16
5. SDK interface specification 16
   5.1. POS terminals information and basic infomation module 16
      5.1.1. Module specification 16
      5.1.2. Interface specification 17
         5.1.2.1. Interface list 17
         5.1.2.2. getDeviceSN 18
         5.1.2.3. getDevicePN 18
         5.1.2.4. getDeviceKSN 18
         5.1.2.5. getDeviceInformation 18
         5.1.2.6. settingsConfiguration 19
         5.1.2.7. installApp 20
         5.1.2.8. uninstallApp 20
         5.1.2.9. setSystemClock 20
         5.1.2.10. reboot 20
         5.1.2.11. recovery 20
         5.1.2.12. setAPN 21
         5.1.2.13. disableStatusBar 21
         5.1.2.14. setLogLevel 21
         5.1.2.15. writeCustomData 22
         5.1.2.16. readCustomData 22
         5.1.2.17. transceive 22
         5.1.2.18. transceive 23
         5.1.2.19. setProperties 23
         5.1.2.20. getProperties 23
         5.1.2.21. setCcidListener 23
         5.1.2.22. isForeground 23
         5.1.2.23. switchAppEnable 24

5.1.2.24. setLauncherApp 24
5.1.2.25. switchGPS 24
5.1.2.26. startLogcat 24
5.1.2.27. stopLogcat 25
5.1.2.28. isDebugMode 25
5.1.2.29. lockDevice 25
5.1.2.30. unlockDevice 25
5.1.2.31. clearAppCache 26
5.1.2.32. setLanguage 26
5.1.3. TYCcidListener 26
5.1.3.1. Interface list 26
5.1.3.2. onCcidReceived 26
5.2. Upgrade module 27
5.1.4. Module description 27
5.1.5. Interface specification 27
5.1.5.1. Interface list 27
5.1.5.2. upgradeFirmware 27
5.1.5.3. upgradeAndroidOS 27
5.1.6. FirmwareUpgradeListener 28
5.1.6.1. Interface list 28
5.1.6.2. onStart 28
5.1.6.3. onProgress 28
5.1.6.4. onError 28
5.1.6.5. onUpdateMainAppSuccess 29
5.1.6.6. onUpdateChildAppSuccess 29
5.1.6.7. onMainAppNewest 29
5.1.6.8. onChildAppNewest 29
5.3. Printing operation module 30
5.1.7. Module specification 30
5.1.8. Interface specification 30
5.1.8.1. Interface list 30
5.1.8.2. initPrinter 31
5.1.8.3. releasePrinter 31
5.1.8.4. feedPaper 31
5.1.8.5. getPrinterStatus 31
5.1.8.6. printText 32
5.1.8.7. printBitmap 32
5.1.8.8. printBarcode 32

5.1.8.9. appendText....32
5.1.8.10. appendBitmap....33
5.1.8.11. appendBarcode....33
5.1.8.12. appendScript....33
5.1.8.13. startPrint....33
5.1.8.14. setTypeface....34
5.1.8.15. setLineSpace....34
5.1.8.16. appendPrintElement....34
5.1.8.17. startPrintElement....34
5.1.9. PrintInitListener....35
5.1.9.1. Interface list....35
5.1.9.2. onPrinterInit....35
5.1.10. PrinterListener....35
5.1.10.1. Interface list....35
5.1.10.2. onPrinterStart....36
5.1.10.3. onPrinterEnd....36
5.1.10.4. onPrinterOutOfPaper....36
5.1.10.5. onPrinterError....36
5.2. Pinpad module....36
5.2.1. Module introduction....36
5.2.2. Interface specification....37
5.2.2.1. Interface list....37
5.2.2.2. updateMainKey....38
5.2.2.3. updateWorkKey....38
5.2.2.4. updateTransKey....38
5.2.2.5. setEnAlgorithm....38
5.2.2.6. calculateMac....39
5.2.2.7. getRandom....39
5.2.2.8. startPinPad....39
5.2.2.9. startPinPad....40
5.2.2.10. cancelPinPad....40
5.2.2.11. selectKeyGroup....41
5.2.2.12. encryptData....41
5.2.2.13. encryptKSN....42
5.2.2.14. updateTransKey(with index)....42
5.2.2.15. updateMainKey(with index)....42
5.2.2.16. updateMainKey(with index)....42
5.2.2.17. updateWorkKey(with index)....43

5.2.2.18. updateWorkKey(with index) 43
5.2.2.19. encryptData(with index) 43
5.2.2.20. calculateMac(with index) 44
5.2.2.21. dukptUpdateKey(with index) 44
5.2.2.22. dukptKsnIncrease(with index) 44
5.2.2.23. dukptGetKsn(with index) 45
5.2.2.24. dukptEncrypt(with index) 45
5.2.2.25. setCipherMode 45
5.2.2.26. setInitVector 46
5.2.3. PinPadResultListener 46
5.2.3.1. Interface list 46
5.2.3.2. onKeyDown 46
5.2.3.3. onGetPinPadResult 46
5.3. Card read module 47
5.3.1. Module specification 47
5.3.2. Bank card interface specification 47
5.3.2.1. Interface list 47
5.3.2.2. setReadCardConfig 48
5.3.2.3. readCard 48
5.3.2.4. cancelReadCard 48
5.3.2.5. startSearchCard 49
5.3.2.6. getAtr 49
5.3.2.7. stopSearchCard 49
5.3.2.8. getMagCardData 50
5.3.2.9. isCardExists 50
5.3.2.10. transmitIC 50
5.3.2.11. transmitRF 50
5.3.3. PSAM Card interface specification 51
5.3.3.1. Interface list 51
5.3.3.2. openPSAMCard 51
5.3.3.3. closePSAMCard 51
5.3.3.4. transmitPSAM 51
5.3.4. Mifare Card interface specification 52
5.3.4.1. Interface list 52
5.3.4.2. verifyKeyA 52
5.3.4.3. verifyKeyB 53
5.3.4.4. readBlock 53
5.3.4.5. writeBlock 53

5.3.4.6. writeValue 53
5.3.4.7. readValue 54
5.3.4.8. increaseValue 54
5.3.4.9. decreaseValue 54
5.3.4.10. readMifareUltralightCard 54
5.3.4.11. writeMifareUltralightCard 55
5.6. EMV process operation module 55
5.3.5. Module introduction 55
5.3.6. Interface specification 55
5.3.6.1. Interface list 55
5.3.6.2. setTermConfig 56
5.3.6.3. getTermConfig 56
5.3.6.4. clearAID 57
5.3.6.5. clearIcAID 57
5.3.6.6. clearRfAID 57
5.3.6.7. clearCAPK 57
5.3.6.8. updateAID 58
5.3.6.9. updateIcAID 58
5.3.6.10. updateRfAID 58
5.3.6.11. updateCAPK 58
5.3.6.12. getAID 59
5.3.6.13. getIcAID 59
5.3.6.14. getRfAID 59
5.3.6.15. getCAPK 59
5.3.6.16. emvProcess 60
5.3.6.17. tradeResponse 60
5.3.6.18. emvCancel 60
5.3.6.19. getEmvHashCode 60
5.3.6.20. setTlv 61
5.3.6.21. getTlv 61
5.3.6.22. getTlvValue 61
5.3.6.23. getTlvList 61
5.3.6.24. getTlvEncrypted 61
5.3.6.25. getTlvValueEncrypted 62
5.3.6.26. getTlvListEncrypted 63
5.3.6.27. queryECBalance 63
5.3.6.28. getEmvCardLog 64
5.3.7. EMVTransListener 64

5.3.7.1. Interface list 64
5.3.7.2. onWaitAppSelect 64
5.3.7.3. onInputAmount 65
5.3.7.4. onConfirmEC 65
5.3.7.5. onConfirmCardNo 65
5.3.7.6. onCertVerfiy 65
5.3.7.7. onCardHolderPwd 66
5.3.7.8. onEmvTerGetInputDisp 66
5.3.7.9. onOffLinePinVerify 66
5.3.7.10. onOffLinePinLastTimeVerify 66
5.3.7.11. onTransResult 67
5.3.8. Parameter Description 67
5.3.8.1. AID/RID Parameter 67
5.3.8.2. CAPK Parameter 68
5.7. Scanner module 68
5.3.9. Module introduction 68
5.3.10. Interface specification 68
5.3.10.1. Interface list 68
5.3.10.2. startScan 68
5.3.10.3. stopScan 69
5.3.11. ScanResultListener 69
5.3.11.1. Interface list 69
5.3.11.2. onGetScanResult 69
5.4. Led module 70
5.4.1. Module introduction 70
5.4.2. Interface specification 70
5.4.2.1. Interface list 70
5.4.2.2. setLed 70
5.5. Beeper module 70
5.5.1. Module introduction 70
5.5.2. Interface specification 71
5.5.2.1. Interface list 71
5.5.2.2. beep 71
5.5.2.3. beep 71
5.5.2.4. beep 71
5.5.2.5. releaseBeeper 71
5.6. Tag module 72
5.6.1. Module introduction 72

5.6.2. Interface specification 72
5.6.2.1. Interface list 72
5.6.2.2. writeTagData 72
5.6.2.3. readTagData 72
5.6.2.4. verifyTag 73
6. Entity class 73
6.1. Introduction 73
6.2. Class list 73
6.3. OperationResult 73
6.4. CardInfo 74
6.5. EMVCardLog 74
6.6. EMVTermConfig 75
6.7. EMVTransData 75
6.8. OnlineResult 76
6.9. PINResult 76
6.10. PinPadResult 76
6.11. Balance 76
6.12. TransResult 76
6.13. DeviceInfo 77
6.14. DeviceVersion 77
6.15. ScanResult 78
6.16. PrintElement 78
6.17. KeyInfo 79
7. status code 79
7.1. Card reading status code(cardInfo.getResponseStatus) 79
7.2. Pin pad status code(pinPadResult.getResultCode) 80
7.3. Firmware upgrade status code(FirmwareUpgradeConstant) 80
7.4. Operation result status code(operationResult.getStatusCode) 81
Appendix 1 Print Instruction 82
1. Initialization 82
2. Print example (recommended) 82
3. Print script format specification 83
3.1 Command overview 83
3.2 Instructions 90
3.3 Output 90
3.4 Sample script 90
Appendix 2 Read card example 93
4. Call the SDK card reader interface 93

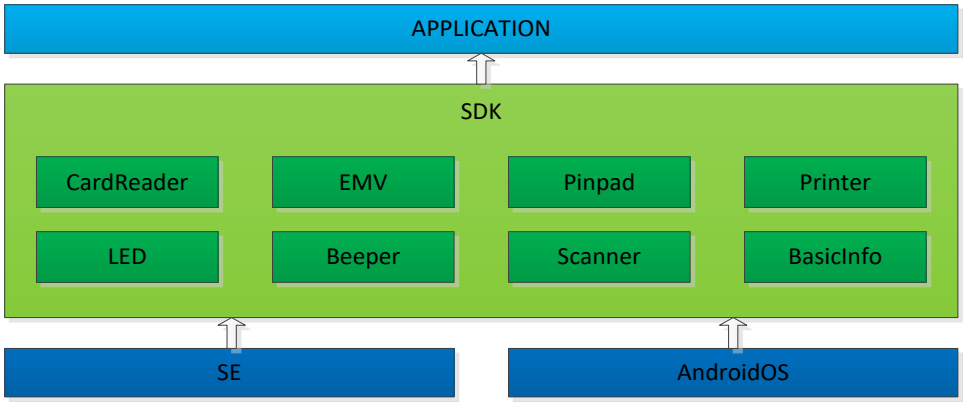
5. User customized card reading 94
2.1 Magnetic stripe cards 94
Appendix 3 Function workflow 98
1. UML diagram 98
1.1 SDK class diagram 98
1.2 Demo class diagram 99
2. Sequence diagram 101
2.1 Card reading sequence diagram 101
2.2 Printing sequence diagram 101
3. Flow chart 102
Appendix 4 Abbreviations 103

1. Purpose

The intelligent POS unified interface query document is provided to facilitate the developers to query the interface, improve the POS terminal software development efficiency, and strengthen the management of intelligent POS terminal software development.

The application in the SDK file consists of eight modules: card reading module、EMV module, pinpad module, printer module, LED control module, beeper module, scanner module and basic info module。

2. Framework



3. Integration instructions

3.1. Development environment

Android Studio : Version 2.0 or higher

JDK : 1.8

```
minSdkVersion 23
targetSdkVersion 29
```

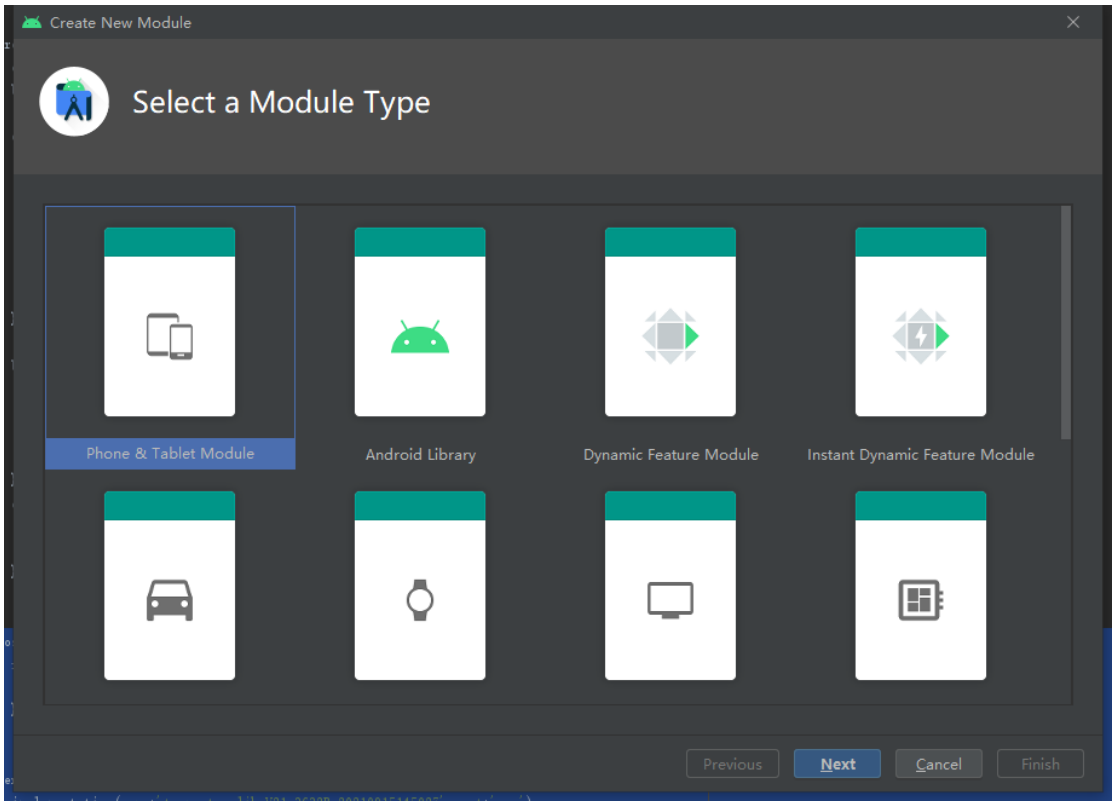
3.2. SDK files

File	Description
tysmartposlib_Vxx_xxxxR_yyyyMMddHHmmss.aar	aar package of API interface

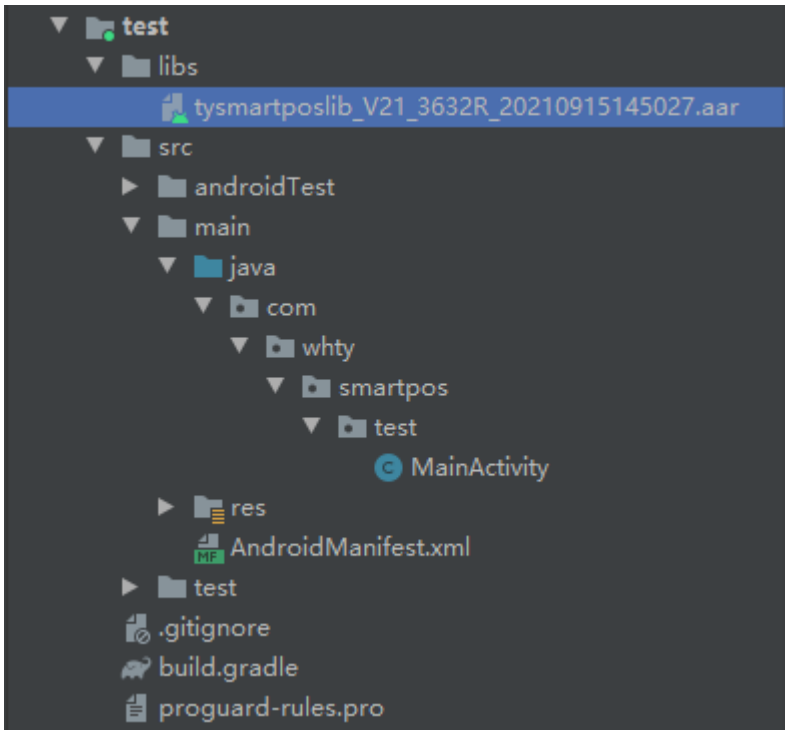
Notes: Vxx is SDK version; xxxx is code base version; R is release version; yyyyMMddHHmmss is building time

3.3. Import SDK

1) Create a new project or open an existing project in AndroidStudio



2) Copy aar to the libs folder in the project



3) Configure build.gradle

```
repositories {
    flatDir {
        dirs 'libs'
    }
}

dependencies {
    implementation(name:'tysmartposlib_V21_3632R_20210915145027',
ext:'aar')
}
```

4) Get API instance

```
import com.whty.smartpos.tysmartposapi.ITYSmartPosApi;
ITYSmartPosApi tyApi = ITYSmartPosApi.get(getApplicationContext());
```

4. SDK description

Define the middle layer interface between application software and smart POS terminals.

The SDK document aims to shield the differences between the applications on Android OS and smart POS terminals, integrate basic operation of function implementation, provide API query. It is convenient for developers to develop customized APPs and improve the efficiency of developers.

The SDK will be output in the form of aar package, and the apk application refers to this aar package and calls it.

5. SDK interface specification

5.1. POS terminals information and basic infomation module

5.1.1. Module specification

This module is mainly to acquire the basic information of the device, such as device SN, PN, KSN, version information, terminal information, etc. It also provides basic operations for some devices, such as APP install and uninstall, system clock setting, APN setting, device restart and other related operations.

5.1.2. Interface specification

5.1.2.1. Interface list

Interface prototype	Interface introduction
getDeviceSN	Get device SN
getDevicePN	Get device PN
getDeviceKSN	Get device KSN
getDeviceInformation	Get device information
settingsConfiguration	Provide some setting switches
installApp	Install APP
uninstallApp	Uninstall APP
setSystemClock	Set System Clock
reboot	Reboot device
recovery	recovery
setAPN	Set APN
disableStatusBar	Hide status bar
setLogLevel	Set Log Level
writeCustomData	Write custom data
readCustomData	Read custom data
transceive	Transceive interface
setProperties	Set Property
getProperties	Get Property
setCcidListener	Set CcidListener
isForeground	Whether the app is in the foreground
switchAppEnable	Disable application switch
setLauncherApp	Set Launcher APP
switchGPS	GPS switch
startLogcat	Start to grab the log

stopLogcat	End grabbing logs
isDebugMode	Whether it is debug mode
lockDevice	Lock the device
unlockDevice	Unlock the device
clearAppCache	Used to clear the cache data of the APPs
setLanguage	Set system language

5.1.2.2. getDeviceSN

Function	String getDeviceSN()	
Description	Get external card reader device SN number(Serial Number), which is the device unique identity	
Parameter	N/A	
Return	String	Return device SN
Instruction	The SN number on the equipment is the device serial number	

5.1.2.3. getDevicePN

Function	String getDevicePN()	
Description	Get external card reader device PN number	
Parameter	N/A	
Return	String	Return device PN
Instruction	N/A	

5.1.2.4. getDeviceKSN

Function	String getDeviceKSN()	
Description	Get device KSN (UnionPay terminal serial number)	
Parameter	N/A	
Return	String	Return device KSN
Instruction	N/A	

5.1.2.5. getDeviceInformation

Function	DeviceInformation getDeviceInformation()
Description	Get device information, including basic device information, device version information, battery information, memory information, storage information, network information, application information, traffic information, etc.

Parameter	N/A	
Return	DeviceInformation	After getting the DeviceInformation, you can get the corresponding information by the following interface: DeviceInfo getDeviceInfo() DeviceVersion getDeviceVersion() BatteryInfo getBatteryInfo() MemoryInfo getMemoryInfo() StorageInfo getStorageInfo() CpuInfo getCpuInfo() NetworkInfo getNetworkInfo() PrinterInfo getPrinterInfo() List<AppInfo> getAppInfoList() TrafficInfo getTrafficInfo(long startTime, long endTime)
Instruction	N/A	

5.1.2.6. settingsConfiguration

Function	boolean settingsConfiguration(SettingsOptions options)		
Description	Provide some setting switches		
Parameter	SettingsOptions	options	Options that need to be set can be added, used as follows: SettingsOptions so = SettingsOptions.create() .setUsbDebugging(true) .setStayAwake(false) .setShowTouchOperation(false) .setPointerLocation(false) .setForceGpuRendering(false) .setEnableStrictMode(false) .setShowAllNonResponsiveApps(true) .setShowPassword(false) .setTimezoneAutomatically(true) .setEnableLocationInformation(false) .setEnableWifi(true) .setEnableBluetooth(false) .setMobileDataNetwork(true) .setMobileDataNetworkRoaming(false);
Return	boolean	true	Set success
		false	Set fail
Instruction	N/A		

5.1.2.7. installApp

Function	boolean installApp(String filePath)		
Description	Install APP，Install app by directory		
Parameter	String	filePath	Apk file path is normally placed under sdcard
Return	boolean	true	APP install success
		false	APP install fail
Instruction	N/A		

5.1.2.8. uninstallApp

Function	boolean uninstallApp(String packageName)		
Description	Uninstall APP, used to uninstall the app on the device		
Parameter	String	packageName	APP packageName
Return	boolean	true	APP uninstall success
		false	APP uninstall fail
Instruction	N/A		

5.1.2.9. setSystemClock

Function	boolean setSystemClock(String time)		
Description	Set the system time, used to set the system time on the device		
Parameter	String	time	Date and; time，format is yyyyMMddhhmmss
Return	boolean	true	Set system time success
		false	Set system time fail
Instruction	N/A		

5.1.2.10. reboot

Function	void reboot()
Description	Restart the device and the device will enter the restart process
Parameter	N/A
Return	N/A
Instruction	N/A

5.1.2.11. recovery

Function	void recovery()
Description	After the interface called, the equipment will automatically enter the process of restoring factory settings

Parameter	N/A
Return	N/A
Instruction	N/A

5.1.2.12. setAPN

Function	boolean setAPN(String name, String apn, String userName, String password)		
Description	set APN according to APN name, ap/username/password to be set		
Parameter	String	name	APN name, which can be defined
	String	apn	ap to be set
	String	userName	username
	String	password	password
Return	boolean	true	Set APN success
		false	Set APN fail
Instruction	The username and password should be passed in when required by the operator, otherwise the value null should be passed in.		

5.1.2.13. disableStatusBar

Function	boolean disableStatusBar(int[] what)		
Description	hide corresponding function keys according to the parameter values passed in		
Parameter	int[]	what	Function key array to hide , defined in PosConstan.StatusBar: DISABLE_NONE: Do not hide button STATUS_BAR_DISABLE_HOME: Hide HOME key STATUS_BAR_DISABLE_BACK: Hide return key STATUS_BAR_DISABLE_RECENT: Hide menu key
Return	boolean	true	Hide status bar success
		false	Hide status bar fail
Instruction	N/A		

5.1.2.14. setLogLevel

Function	void setLogLevel(int level)		
Description	Set log levels according to the parameter values input		
Parameter	int	level	Log level, defined in PosConstant.LogLevel: VERBOSE: verbose DEBUG: debug INFO: info ERROR: error
Return	N/A		

Instruction	N/A

5.1.2.15. writeCustomData

Function	int writeCustomData(int position, String data)		
Description	Write customized data into the device according to the position and customized data input		
Parameter	int	position	Custom data location, defined in PosConstant.customData: CUSTOM_DATA_FIRST: Customized information 1 CUSTOM_DATA_SECOND: Customized information 2
	String	data	Data to be written
Return	int	0	Write data success
		not 0	Write data fail
Instruction	The data is stored in the security chip and can be used to store some important data, with a length of no more than 256 bytes		

5.1.2.16. readCustomData

Function	OperationResult readCustomData(int position);		
Description	Read customized data from the device according to the position of customized data passed in		
Parameter	int	position	Custom data position, defined in PosConstant.customData: CUSTOM_DATA_FIRST: Custom information 1 CUSTOM_DATA_SECOND: Custom information 2
Return	OperationResult	0	The custom information is successfully written by getData()
		Not 0	Fail
Instruction	When the customized data is null, the status code returns 0x6f00		

5.1.2.17. transceive

Function	OperationResult transceive(String cmd)		
Description	Transceive interface, Send command string according to the parameters passed in		
Parameter	String	cmd	Transceive command
Return	OperationResult	0	Success, get the returned data by getData()
		Not 0	Fail
Instruction	N/A		

5.1.2.18. transceive

Function	byte[] transceive(byte[] cmd)		
Description	Transceive interface，Send command bytes according to the parameters passed in		
Parameter	byte[]	cmd	Transceive command
Return	byte[]	Byte data returned	
Instruction	N/A		

5.1.2.19. setProperties

Function	boolean setProperties(String key, String value)		
Description	Set system properties according to the keys and values passed in		
Parameter	String	key	Key
	String	value	Value
Return	boolean	true	Set properties success
		false	Set properties fail
Instruction	Used to set modifiable system properties such as persist.sys.tphprinter.grade		

5.1.2.20. getProperties

Function	String getProperties(String key, String defaultValue)		
Description	get system properties according to key		
Parameter	String	key	Key
	String	defaultValue	Default value
Return	String	Return system attribute data	
Instruction	N/A		

5.1.2.21. setCcidListener

Function	void setCcidListener(TYCcidListener ccidListener)		
Description	Set ccid listener		
Parameter	TYCcidListener	ccidListener	Referring to 5.1.3 TYCcidListener
Return	N/A		
Instruction	command for external application to proceed f0ae		

5.1.2.22. isForeground

Function	boolean isForeground(String pkgName)
Description	Judge whether the app is in the foreground according to the package name

	passed in		
Parameter	String	pkgName	Package name
Return	boolean	true	APP in the foreground
		false	APP not in the foreground
Instruction	N/A		

5.1.2.23. switchAppEnable

Function	boolean switchAppEnable(String pkgName, boolean enable)		
Description	Enable or disable app according to the parameter value passed in		
Parameter	String	pkgName	Package name
	String	enable	Whether enable
Return	boolean	true	Operation successful
		false	Operation failed
Instruction	N/A		

5.1.2.24. setLauncherApp

Function	boolean setLauncherApp(String pkgName,String className)		
Description	Set the launcher apps according to the package name and class name passed in		
Parameter	String	pkgName	Package name
	String	className	Class name
Return	boolean	true	Operation successful
		false	Operation failed
Instruction	N/A		

5.1.2.25. switchGPS

Function	boolean switchGPS(boolean enable)		
Description	set GPS switch according to the input parameter value		
Parameter	boolean	enable	swich GPS on/off
Return	boolean	true	Set successfully
		false	Set failed
Instruction	N/A		

5.1.2.26. startLogcat

Function	boolean startLogcat(LogcatFilter logcatFilter)		
Description	Start to grab the log, you can set the filter condition with LogcatFilter		
Parameter	LogcatFilt	logcatFilte	Conditional filter, used as follows:

	er	r	LogcatFilter logcatFilter = LogcatFilter.create() .addCondition(new LogcatFilter.Package("xxx")) .addCondition(new LogcatFilter.Level("v")) .setTimeout(10 * 60 * 1000) .setRemote(true);
Return	boolean	true	Start successfully
		false	Start failed
Instruction	Need to call stopLogcat to end grabbing logs		

5.1.2.27. stopLogcat

Function	String stopLogcat(boolean needUpload)		
Description	End grabbing logs, and decide whether to upload logs according to parameter		
Parameter	boolean	needUpload	Upload the log or delete this log
Return	String	path of log stored	
Instruction	When calling this interface, the parameter is passed in true, and the saved log will be automatically uploaded to the TMS server		

5.1.2.28. isDebugMode

Function	boolean isDebugMode()		
Description	Judge whether the equipment is in debugging mode		
Parameter	N/A		
Return	boolean	true	Debug mode
		false	Not debug mode
Instruction	N/A		

5.1.2.29. lockDevice

Function	boolean lockDevice(String message)		
Description	Lock the device		
Parameter	String	message	Prompt text
Return	boolean	true	Lock successfully
		false	Lock failed
Instruction	Call unlockDevice to unlock the device		

5.1.2.30. unlockDevice

Function	boolean unlockDevice()
Description	Unlock the device

Parameter	N/A		
Return	boolean	true	Unlock successfully
		false	Unlock failed
Instruction	Call unlockDevice to unlock the device		

5.1.2.31. clearAppCache

Function	boolean clearAppCache(String[] appPkgNames)		
Description	Used to clear the cache data of the APPs		
Parameter	String[]	appPkgNames	APP package name array
Return	boolean	true	Clear successfully
		false	Clear failed
Instruction	N/A		

5.1.2.32. setLanguage

Function	boolean setLanguage(String... languages)		
Description	Set system language		
Parameter	String...	languages	Language-Country (en-US、zh-CN...)
Return	boolean	true	Set successfully
		false	Set failed
Instruction	N/A		

5.1.3. TYCcidListener

5.1.3.1. Interface list

Interface prototype	**Interface description**
onCcidReceived	Receive CCID command

5.1.3.2. onCcidReceived

Function	byte[] onCcidReceived(byte p1, byte p2, byte[] data)		
Description	command for external application to proceed f0ae and returning data		
Parameter	byte	p1	Instruction P1
	byte	p2	Instruction p2
	byte[]	data	Instruction data
Return	byte[]	Return data after command proceeded	
Instruction	N/A		

5.2. Upgrade module

5.1.4. Module description

This module is used for Android system upgrade and security chip firmware upgrade.

5.1.5. Interface specification

5.1.5.1. Interface list

Interface prototype	**Interface description**
upgradeFirmware	Upgrade Firmware
upgradeAndroidOS	Upgrade AndroidOS

5.1.5.2. upgradeFirmware

Function	void upgradeFirmware(String mainFilePath, String childFilePath, FirmwareUpgradeListener listener)		
Description	Upgrade the firmware of the security chip according to the parameters passed in: main application path, sub application path and status callback interface		
Parameter	String	mainFilePath	Main application path
	String	childFilePath	Sub application path
	FirmwareUpgradeListener	listener	Callback function , Referring to 5.2.3 FirmwareUpgradeListener
Return	N/A		
Instruction	N/A		

5.1.5.3. upgradeAndroidOS

Function	void upgradeAndroidOS(String filePath)		
Description	Upgrade the Android operating system according to the upgrade file path		
Parameter	String	filePath	path of upgrade file，file name is update.zip
Return	N/A		
Instruction	After the OTA upgrade package is detected, the system will automatically restart and enter the upgrade process		

5.1.6. FirmwareUpgradeListener

5.1.6.1. Interface list

Interface prototype	**Interface description**
onStart	Start to upgrade
onProgress	Progress callback
onError	Upgrade error
onUpdateMainAppSuccess	Main application upgrade successfully
onUpdateChildAppSuccess	Sub application upgrade successfully
onMainAppNewest	Main application version is newest
onChildAppNewest	Sub application version is newest

5.1.6.2. onStart

Function	void onStart(byte type)		
Description	The firmware upgrade starts, and the parameters include the application type		
Parameter	byte	type	00 Main application，01 Sub application
Return	N/A		
Instruction	N/A		

5.1.6.3. onProgress

Function	void onProgress(byte type, int progressValue)		
Description	The current progress of upgrading firmware. The parameters include application type and current progress value		
Parameter	byte	type	00，Main application；01，Sub application
	int	progressValue	Progress value，0-100
Return	N/A		
Instruction	N/A		

5.1.6.4. onError

Function	void onError(String code,String msg)
Description	Failed to upgrade firmware. The parameters contain error code and error information

Parameter	String	code	Error code，defined in FirmwareUpgradeConstant
	String	msg	Error message
Return	N/A		
Instruction	N/A		

5.1.6.5. onUpdateMainAppSuccess

Function	void onUpdateMainAppSuccess(String version)		
Description	The main application is updated successfully, and the parameters contain the version information of the main application		
Parameter	String	version	Version message
Return	N/A		
Instruction	N/A		

5.1.6.6. onUpdateChildAppSuccess

Function	void onUpdateChildAppSuccess(String version)		
Description	The sub application is updated successfully, and the parameters contain the sub application version information		
Parameter	String	version	Version message
Return	N/A		
Instruction	N/A		

5.1.6.7. onMainAppNewest

Function	void onMainAppNewest()
Description	The main app version has been upgraded to the latest version
Parameter	N/A
Return	N/A
Instruction	N/A

5.1.6.8. onChildAppNewest

Function	void onChildAppNewest()
Description	The sub app version has been upgraded to the latest version
Parameter	N/A
Return	N/A
Instruction	N/A

5.3. Printing operation module

5.1.7. Module specification

This module mainly provides the interfaces related to the printing function of the device. After successfully initializing the printer and setting the printing parameters, the relevant interfaces can be called. The interfaces provide the following functions, such as printing pictures, printing QR codes, printing strings of text, template printing, script printing, paper header spacing control, etc.

5.1.8. Interface specification

5.1.8.1. Interface list

Interface prototype	**Interface description**
initPrinter	Initialize the printer
getPrinterStatus	Printer status
feedPaper	Feed paper
printText	Print text
printBitmap	Print bitmap
printBarcode	Print barcode
printCustomTemplate	Print custom template
appendText	Add text
appendBitmap	Add bitmap
appendBarcode	Add barcode
appendScript	Add script
startPrint	Print the added print content
setTypeface	Set printing typeface，valid only on PrintElement
setLineSpace	Set printing line space，valid only on PrintElement
appendPrintElement	Append print element

startPrintElement	Print the added PrintElement
releasePrinter	Release printer

5.1.8.2. initPrinter

Function	boolean initPrinter(PrinterInitListener listener)		
Description	Initialize the printer and receive the initialization result in the callback function		
Parameter	PrinterInitListener	listener	Initialize printer listener , referring to 5.3.3 PrinterInitListener
Return	boolean	true	Initialization successful
		false	Initialization failed
Instruction	When receiving the callback, the printer initialization is completed, then printing starts.		

5.1.8.3. releasePrinter

Function	void releasePrinter()
Description	Release printer
Parameter	N/A
Return	N/A
Instruction	the printer needs to reinitialize after release

5.1.8.4. feedPaper

Function	int feedPaper(int value)		
Description	Feed paper		
Parameter	int	value	Paper feeding amount, in pixels
Return	int	0	Print successfully
		Not 0	Print failed
Instruction	N/A		

5.1.8.5. getPrinterStatus

Function	int getPrinterStatus()	
Description	Get the current status of the printer	
Parameter	N/A	
Return	int	Printer status，defined in PrinterConstant. PrintResult： SUCCESS：normal ERROR：abnormal OUT_OF_PAPER： printer out of paper

		LOW_POWER：Low charge OVERHEATED：The temperature is too high PARAMS_ERROR：Parameter error PRINTER_SERVICE_NOT_BOUND：The print service is not binded
Instruction	N/A	

5.1.8.6. printText

Function	int printText(String text, int align, int fontSize, boolean isBold)		
Description	Print text		
Parameter	String	text	Text to be printed
	int	align	Defined in Align
	int	fontSize	Defined int FontSize
	boolean	isBold	Bold or not
Return	int	0	Print successfully
		Not 0	Print failed
Instruction	N/A		

5.1.8.7. printBitmap

Function	int printBitmap(Bitmap bitmap)		
Description	Print bitmap		
Parameter	Bitmap	bitmap	Bitmap to be printed
Return	int	0	Print successfully
		Not 0	Print failed
Instruction	N/A		

5.1.8.8. printBarcode

Function	int printBarcode(int barcodeType, String barcode)		
Description	Print barcode		
Parameter	int	barcodeType	Barcode type, Defined in QrCodeType
	String	barcode	Barcode text
Return	int	0	Print successfully
		Not 0	Print failed
Instruction	N/A		

5.1.8.9. appendText

Function	void appendText(String text, int align, int fontSize, boolean isBold)

Description	Appent text		
Parameter	String	text	Text to be added
	int	align	Defined in Align
	int	fontSize	Defined int FontSize
	boolean	isBold	Bold or not
Return	int	0	Print successfully
		Not 0	Print failed
Instruction	N/A		

5.1.8.10. appendBitmap

Function	void appendBitmap(Bitmap bitmap)		
Description	Append bitmap		
Parameter	Bitmap	bitmap	Bitmap to be added
Return	int	0	Print successfully
		Not 0	Print failed
Instruction	N/A		

5.1.8.11. appendBarcode

Function	void appendBarcode(int barcodeType, String barcode)		
Description	Append barcode		
Parameter	int	barcodeType	Barcode type, Defined in QrCodeType
	String	barcode	Barcode text
Return	int	0	Print successfully
		Not 0	Print failed
Instruction	N/A		

5.1.8.12. appendScript

Function	void appendScript(String text)		
Description	Append print script		
Parameter	String	text	Script data, reference resources Appendix I 3.Print script format specification
Return	N/A		
Instruction	Need to call 'startPrint' function to print		

5.1.8.13. startPrint

Function	void startPrint(PrinterListener printerListener)

Description	Print the added print content by appendText、appendBitmap、appendBarcode、appenScript, and monitor the start/end/abnormal state of printing		
Parameter	PrinterListener	printerListener	Printer listener，referring to 5.3.4 PrinterListener
Return	N/A		
Instruction	N/A		

5.1.8.14. setTypeface

Function	void setTypeface(Typeface tf)		
Description	Set printing font. valid only on PrintElement		
Parameter	Typeface	tf	Printing font. Fonts are stored under the Android directory (/ system / fonts) or imported font resources.
Return	N/A		
Instruction	This interface is applicable to the font printed when ` appendPrintElement ` is called		

5.1.8.15. setLineSpace

Function	void setLineSpace(int yPixel)		
Description	Set the line spacing. valid only to the line spacing printed when ` appendPrintElement ` is called		
Parameter	int	yPixel	Line spacing，In pixels
Return	N/A		
Instruction	This interface is applicable to the line spacing printed when `appendPrintElement` is called		

5.1.8.16. appendPrintElement

Function	void appendPrintElement(PrintElement printElement)		
Description	Append print element		
Parameter	PrintElement	printElement	Print element，referring to 6.16 PrintElement
Return	N/A		
Instruction	N/A		

5.1.8.17. startPrintElement

Function	Int startPrintElement()
Description	Start printing appended printing elements
Parameter	N/A

Return	int	Print results，defined in PrinterConstant.PrintResult： SUCCESS，success ERROR，error OUT_OF_PAPER，Paper shortage LOW_POWER，Low power OVERHEATED，Overheated PARAMS_ERROR，Parameter error PRINTER_SERVICE_NOT_BOUND，Print service not binded (printer not initialized)
Instruction	N/A	

5.1.9. PrintInitListener

5.1.9.1. Interface list

Interface prototype	**Interface description**
onPrinterInit	Printer initialization completion callback

5.1.9.2. onPrinterInit

Function	void onPrinterInit(boolean isSuccess)		
Description	Printer initialization callback		
Parameter	boolean	isSuccess	true，success
			false，fail
Return	N/A		
Instruction	N/A		

5.1.10. PrinterListener

5.1.10.1. Interface list

Interface prototype	**Interface description**
onPrinterStart	Printing starts
onPrinterEnd	Printing complete
onPrinterOutOfPaper	printer out of paper
onPrinterError	Print error

5.1.10.2. onPrinterStart

Function	void onPrinterStart(String msg)		
Description	Printing starts		
Parameter	String	msg	title of the currently printing document
Return	N/A		
Instruction	N/A		

5.1.10.3. onPrinterEnd

Function	void onPrinterEnd(String msg)		
Description	Printing complete		
Parameter	String	msg	description of the currently printed document
Return	N/A		
Instruction	N/A		

5.1.10.4. onPrinterOutOfPaper

Function	void onPrinterOutOfPaper()
Description	Printer out of paper
Parameter	N/A
Return	N/A
Instruction	N/A

5.1.10.5. onPrinterError

Function	void onPrinterError(int errorCode, String errorMsg)		
Description	Print error and return error message		
Parameter	int	errorCode	Error code，defined in PrinterConstant.PrintResult： LOW_POWER：Low power OVERHEATED：Printer overheating ERROR：Printing error
	String	msg	description of printing error
Return	N/A		
Instruction	N/A		

5.2. Pinpad module

5.2.1. Module introduction

This module mainly manages the call of keys and pinpad. Keys management includes three types: updating the main key, working key, and tranfer key. For the

double-length key algorithm, the working key length is 20 bytes, of which the first 16 bytes are the working key and the last 4 bytes are checkValue. CheckValue is the working key plain text. CheckValue is the the first 4 bytes of the result of double key algorithm with working key plain text to 8 values 0x00.

5.2.2. Interface specification

5.2.2.1. Interface list

Interface prototype	**Interface introduction**
updateMainKey	Update main key
updateWorkKey	Update work key
updateTransKey	Update transfer key
setEnAlgorithm	Set encryption and decryption algorithm
calculateMac	Calculate MAC
getRandom	Get random number
startPinPad	Start Pinpad
cancelPinPad	Cancel Pinpad
selectKeyGroup	select key group
encryptData	Encrypt data
encryptKSN	Encrypt KSN
updateTransKey(with index)	Update transfer key
updateMainKey(with index)	Update main key
updateWorkKey(with index)	Update work key
encryptData(with index)	Encrypt data
calculateMac(with index)	Calculate MAC
dukptUpdateKey(with index)	Dukpt update Key
dukptKsnIncrease(with index)	Dukpt KSN increase
dukptGetKsn(with index)	Dukpt get KSN
dukptEncrypt(with index)	Dukpt Encrypt data

setCipherMode	Set cipher mode
setInitVector	Set initial vector

5.2.2.2. updateMainKey

Function	int updateMainKey(String mainKey, String checkValue)		
Description	Update main key		
Parameter	String	mainKey	Key data
	String	checkValue	Check value
Return	int	0	Success
		Not 0	Failed
Instruction	When the transfer key is all F, the master key can be updated by passing null in the check value		

5.2.2.3. updateWorkKey

Function	int updateWorkKey(int type, String key, String checkValue)		
Description	Update work key		
Parameter	int	type	Key type: Defined in PinPadConstant.KeyType: MACKEY: MAC key TDKEY: Track key PINKEY: PIN key
	String	key	Key data
	String	checkValue	Check value
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.2.2.4. updateTransKey

Function	int updateTransKey(String transKey)		
Description	Update transfer key		
Parameter	String	transKey	Transfer secret key plaintext (string length 32)
Return	int	0	Success
		Not 0	Fail
Instruction	for debug only		

5.2.2.5. setEnAlgorithm

Function	int setEnAlgorithm(int type)
Description	Set encryption and decryption algorithm

Parameter	int	type	Encryption and decryption algorithm, defined in PinPadConstant.EnAlgorithm THREE_DES SM4
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.2.2.6. calculateMac

Function	OperationResult calculateMac(byte[] data,int calMode, Object... otherParams)		
Description	Calculate MAC value		
Parameter	byte[]	data	Byte array for calculation
	int	calMode	MAC algorithm, defined in PinPadConstant.MacAlgorithm UNION X99 X919
	Object...	otherParams	Initial vector, which needs to be passed in for CBC algorithm
Return	Operation Result	0	Success, get MAC value by getData()
		Not 0	Failed
Instruction	N/A		

5.2.2.7. getRandom

Function	OperationResult getRandom()		
Description	Obtain the 8-byte random number produced by the device		
Parameter	N/A		
Return	Operation Result	0	Success, get random number by getData()
		Not 0	Failed
Instruction	N/A		

5.2.2.8. startPinPad

Function	void startPinPad(PinPadConfig pinPadConfig, PinPadResultListener listener)		
Description	Start password keyboard		
Parameter	PinPadConfig	pinPadConfig	Set configuration as needed, used as follows: PinPadConfig config = PinPadConfig.create() .setCardNum("6241662312653212") .setAmount(100) .setTimeout(60)

			.setMinLen(4) .setMaxLen(6) .isTurnOnSound(true) .isDisplayBar(true) .setPinFormat(PinPadConstant.PinFormat.FORMAT_0) .setKeyIndex(0);
	PinPadResultListener	listener	Input result callback, detail in 5.4.3 PinPadResultListener
Return	N/A		
Instruction	N/A		

5.2.2.9. startPinPad

Function	void startPinPad(PinPadConfig pinPadConfig, ArrayList<KeyInfo> keyInfos, PinPadResultListener listener)		
Description	Start customized pinpad		
Parameter	PinPadConfig	pinPadConfig	Set configuration as needed, used as follows: PinPadConfig config = PinPadConfig.create() .setCardNum("6241662312653212") .setAmount(100) .setTimeout(60) .setMinLen(4) .setMaxLen(6) .isTurnOnSound(true) .isDisplayBar(true) .setPinFormat(PinPadConstant.PinFormat.FORMAT_0) .setKeyIndex(0);
	ArrayList<KeyInfo>	keyInfos	keys attributes such as key position, width, height, color, referring to 6.17 KeyInfo
	PinPadResultListener	listener	Input result callback, referring to 5.4.3 PinPadResultListener
Return	N/A		
Instruction	N/A		

5.2.2.10. cancelPinPad

Function	void cancelPinPad()
Description	Cancel pinpad entry
Parameter	N/A
Return	N/A

Instruction	N/A

5.2.2.11. selectKeyGroup

Function	int selectKeyGroup(int index)		
Description	Select the key group and set the key index		
Parameter	int	index	Key index（0~9）
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.2.2.12. encryptData

Function	OperationResult encryptData(int keyId, byte[] data, int calMode, byte[]... vector)		
Description	Encrypted data		
Parameter	int	keyId	Specified encryption key id, defined in PinPadConstant.KeyId: KEY_ID_FIRST_MAINKEY KEY_ID_SECOND_MAINKEY KEY_ID_FIRST_TDKEY KEY_ID_FIRST_PINKEY KEY_ID_FIRST_MACKEY DUKPT group key id，defined in PinPadConstant.Dukpt： DUKPT_PIN_KEY DUKPT_MAC_KEY_REQ DUKPT_MAC_KEY_RES DUKPT_DAT_KEY_REQ DUKPT_DAT_KEY_RES
	byte[]	data	Encrypted data
	int	calMode	Encryption or decryption, defined in PinPadConstant.EncryptMode： ENCRYPT DECRYPT DUKPT_ENCRYPT DUKPT_DECRYPT
	byte[]...	vector	Vector, which when passed in represents the use of CBC encryption
Return	Operation Result	0	Success，get the encryption result by getData()
		Not 0	Failed
Instruction	N/A		

5.2.2.13. encryptKSN

Function	OperationResult encryptKSN(String KSN, String scatterFactor)		
Description	Encrypted KSN (UnionPay terminal serial number)		
Parameter	String	KSN	KSN
	String	scatterFactor	Encryption factor,normally the last 6 digits of the card number
Return	Operation Result	0	Success，get the encrypted KSN by getData()
		Not 0	Failed
Instruction	N/A		

5.2.2.14. updateTransKey(with index)

Function	int updateTransKey(int transferKeyIndex, String clearTransferKey)		
Description	Update clear trans key		
Parameter	int	transferKey Index	transferKeyIndex index
	String	clearTransferKey	Clear key data
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.2.2.15. updateMainKey(with index)

Function	int updateMainKey(int transferKeyIndex, int mainKeyIndex, String mainKey, String checkValue)		
Description	Update ciphertext main key		
Parameter	int	transferKey Index	Transfer key index
	int	mainKeyIndex	Main key index
	String	mainKey	Ciphertext main key
	String	checkValue	Check value
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.2.2.16. updateMainKey(with index)

Function	(int mainKeyIndex, String mainKey)

Description	Update clear main key		
Parameter	String	mainKeyIndex	Key index
	String	mainKey	Clear key data
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.2.2.17. updateWorkKey(with index)

Function	int updateWorkKey(int mainKeyIndex, int workKeyIndex, String workKey, String checkValue)		
Description	Update ciphertext work key		
Parameter	int	mainKeyIndex	Main key index
	int	workKeyIndex	Work key index
	String	workKey	Ciphertext work key
	String	checkValue	Check value
Return	int	0	Success
		Not 0	Failed
Instruction	Check value supports 4 bytes, 3 bytes and null		

5.2.2.18. updateWorkKey(with index)

Function	updateWorkKey(int workKeyIndex, String workKey)		
Description	Update clear work key		
Parameter	String	workKeyIndex	Key index
	String	workKey	Clear key data
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.2.2.19. encryptData(with index)

Function	OperationResult encryptData(int keyIndex, int calMode, byte[] data)		
Description	Encrypt data		
Parameter	int	keyIndex	Key index used for encryption
	int	calMode	Encryption or decryption, defined in PinPadConstant.EncryptMode: ENCRYPT

			DECRYPT
	byte[]	data	Encrypted data
Return	Operation Result	0	Success，get the encryption result by getData()
		Not 0	Failed
Instruction	N/A		

5.2.2.20. calculateMac(with index)

Function	OperationResult calculateMac(int keyIndex, byte[] data, int algorithm)		
Description	Calculate MAC value		
Parameter	int	keyIndex	Key index used for computation
	byte[]	data	For calculating data
	int	algorithm	Algorithm, defined in PinPadConstant.MacAlgorithm： UNION X99 X919
Return	Operation Result	0	Success，get the MAC data by getData()
		Not 0	Failed
Instruction	N/A		

5.2.2.21. dukptUpdateKey(with index)

Function	int dukptUpdateKey(int transferKeyIndex, int keyIndex, int keyType, String keyData, String ksn)		
Description	DUKPT update BDK or IPEK		
Parameter	int	transferKey Index	Transfer key index
	int	keyIndex	Key index
	int	keyType	Key type, defined in PinPadConstant.DukptKeyType： BDK IPEK
	String	keyData	Key data
	String	ksn	Ksn, 10 bytes
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.2.2.22. dukptKsnIncrease(with index)

Function	int dukptKsnIncrease(int keyIndex)
Description	KSN Increase

Parameter	int	keyIndex	Key index
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.2.2.23. dukptGetKsn(with index)

Function	OperationResult dukptGetKsn(int keyIndex)		
Description	Get current ksn		
Parameter	int	keyIndex	Key index
Return	Operation Result	0	Success，get the ksn by getData()
		Not 0	Failed
Instruction	N/A		

5.2.2.24. dukptEncrypt(with index)

Function	OperationResult dukptEncrypt(int keyIndex, int calMode, byte[] data)		
Description	DUKPT encrypt data		
Parameter	int	keyIndex	Key index
	int	calMode	Encryption and decryption method, defined in PinPadConstant.DukptKeyMode： REQUEST_ENCRYPT REQUEST_DECRYPT RESPONSE_ENCRYPT RESPONSE_DECRYPT
	byte[]	data	Encrypted data
Return	Operation Result	0	Success，get the encryption result by getData()
		Not 0	Failed
Instruction	N/A		

5.2.2.25. setCipherMode

Function	int setCipherMode(int cipherMode)		
Description	Set ECB or CBC		
Parameter	int	cipherMode	Cipher mode, defined in PinPadConstant.CipherMode： ECB CBC
Return	int	0	Success
		Not 0	Failed
Instruction	Default is ECB		

5.2.2.26. setInitVector

Function	int setInitVector(byte[] iv)		
Description	Set initial vector		
Parameter	byte[]	iv	initial vector
Return	int	0	Success
		Not 0	Failed
Instruction	When using CBC encryption, need to set the initial vector, the default is all 0		

5.2.3. PinPadResultListener

5.2.3.1. Interface list

Interface prototype	**Interface description**
onKeyDown	Inputting
onGetPinPadResult	Input completed

5.2.3.2. onKeyDown

Function	void onKeyDown(int count)		
Description	pinpad click function key callback		
Parameter	int	count	Number of digits of the current password inputting
Return	N/A		
Instruction	N/A		

5.2.3.3. onGetPinPadResult

Function	void onGetPinPadResult(PinPadResult pinPadResult)		
Description	Password keyboard result callback		
Parameter	PinPadResult	pinPadResult	Password input result, detail in 6.10 PinPadResult resultCode: statuscode, defined in PinPadConstant.PinPad: SUCCESS: Success BYPASS: confirmation without password CANCEL: Cancel TIMEOUT: Timeout ERROR: Error pinBlock: Encrypted pinBlock
Return	N/A		

Instruction	N/A

5.3. Card read module

5.3.1. Module specification

Card read module mainly reads card information including the following two methods :

1. After successfully setting the card reader configuration , call readCard method to get card information (magnetic stripe card and IC card) or call cancelReadCard method to cancel the card read operation.
2. Start the card search operation. call startSearchCard method, then call getMagCardData after the card then get magnetic strip card information found. Start EMV process for getting IC card information and call stopSearchCard method to cancel the card read operation.

5.3.2. Bank card interface specification

5.3.2.1. Interface list

Interface prototype	**Interface description**
setReadCardConfig	Set card reading configuration
readCard	Card reading operation，with the Call of setReadCardConfig
cancelReadCard	Cancel card reading operation
startSearchCard	Start card search operation
getAtr	Get card ATR
stopSearchCard	Stop card search operation
getMagCardData	Get magnetic strip card info
isCardExists	Is IC card in place
transmitIC	IC card transceive interface

transmitRF	Contactless card transceive interface

5.3.2.2. setReadCardConfig

Function	void setReadCardConfig(boolean msr, boolean ic, boolean rf)		
Description	Set the card reading configuration of switching magnetic stripe card/contact card /contactless card on or off		
Parameter	boolean	msr	Open magnetic stripe card channel
	boolean	ic	Open contact card channel
	boolean	rf	Open contactless card channel
Return	N/A		
Instruction	N/A		

5.3.2.3. readCard

Function	CardInfo readCard(String amount, String tradeTime, byte tradeType,byte timeout, Object... otherParams)		
Description	The card read default interface to get card data after after setReadCardConfig called		
Parameter	String	amount	Amount, the length of the incoming string should not exceed 12 bits
	String	tradeTime	Trading time，format "MMddHHmmss"
	byte	tradeType	Transaction type (0x00 represents consumption, 0x31 represents balance query)
	byte	timeout	Timeout, hexadecimal
	Object...	otherParams	Other parameters (random number, supporting byte [] and string types). Normally this parameter does not need to be passed in
Return	CardInfo	Card info，referring to 6.4 CardInfo	
Instruction	This interface is only used to get card info.		

5.3.2.4. cancelReadCard

Function	boolean cancelReadCard()		
Description	Canceling card reading is only applicable to canceling after calling readcard		
Parameter	N/A		
Return	boolean	true	Cancel success
		false	Cancel fail

Instruction	N/A

5.3.2.5. startSearchCard

Function	int startSearchCard(boolean msr, boolean ic, boolean rf, int timeout);		
Description	Start card searching. This interface is only used to detect the status of cards		
Parameter	boolean	msr	Magnetic stripe card channel
	boolean	ic	Contact card channel
	boolean	rf	Non card receiving channel
	int	timeout	Timeout
Return	int	Card type is defined in CardReaderConstant.CardStatus: SEARCH_CARD_MAG: Magnetic stripe card detected SEARCH_CARD_MAG_ERROR: Magnetic stripe card detected, but without data of track two SEARCH_CARD_IC_IN: IC card insertion detected SEARCH_CARD_IC_OUT: IC card removal detected SEARCH_CARD_IC_ERROR: IC card insertion is detected, but the card is abnormal SEARCH_CARD_RF_IN: contactless card insertion detected SEARCH_CARD_RF_OUT: Non card removal detected SEARCH_CARD_TIME_OUT: Card search timeout SEARCH_CARD_CANCEL: Card search cancel	
Instruction	N/A		

5.3.2.6. getAtr

Function	String getAtr()	
Description	Get card ATR	
Parameter	N/A	
Return	String	The ATR of IC card
Instruction	The contactless card has atr and uid, the contact card only has atr, the format is TLV, the tag of ATR is 01, and the tag of UID is 02. e.g. 01143b8f80010031c173c82110645749303600900098020 4bf17f412	

5.3.2.7. stopSearchCard

Function	int stopSearchCard()		
Description	Stop card searching, which is only applicable to the cancellation operation after startSearchCard called		
Parameter	N/A		
Return	int	0	successful

		Not 0	failed
Instruction	N/A		

5.3.2.8. getMagCardData

Function	CardInfo getMagCardData(Object...otherParams)		
Description	get magnetic stripe card information		
Parameter	Object...	otherParams	Other parameters (random number, supporting byte [] and string types). Normally the parameter does not need to be passed in
Return	CardInfo	Magnetic stripe card information，referring to 6.4 CardInfo	
Instruction	This interface needs to be called after startSearchCard is successful.		

5.3.2.9. isCardExists

Function	int isCardExists()		
Description	judge whether the IC card is in place		
Parameter	N/A		
Return	int	0	IC card is in place
		Not 0	IC card is not in place
Instruction	N/A		

5.3.2.10. transmitIC

Function	OperationResult transmitIC(byte[] data)		
Description	Contact card data transceive interface		
Parameter	byte[]	data	Byte array，Instruction data field
Return	OperationResult	0	Success，get the returned data by getData()
		Not 0	Fail
Instruction	the interface should be called after the contact card channel is opened, otherwise the status code 0x6F00 will be returned.		

5.3.2.11. transmitRF

Function	OperationResult transmitRF(byte[] data)		
Description	Non card data transceive interface		
Parameter	byte[]	data	Byte array，data field of command
Return	OperationResult	0	Success，get the returned data by getData()
		Not 0	Failed
Instruction	the interface should be called after the contact card channel is opened, otherwise the status code 0x6F00 will be returned.		

5.3.3. PSAM Card interface specification

5.3.3.1. Interface list

Interface prototype	**Interface description**
openPSAMCard	Open the PSAM card channel
closePSAMCard	Close the PSAM card channel
transmitPSAM	PSAM card transmit

5.3.3.2. openPSAMCard

Function	int openPSAMCard(int id)		
Description	Open the PSAM card channel and call this interface before transparent transmission of PSAM card		
Parameter	int	id	Card slot id，defined in CardReaderConstant.Psam: PSAM_CARD_1：Card channel 1 PSAM_CARD_2：Card channel 2 PSAM_CARD_3：Card channel 3
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.3.3.3. closePSAMCard

Function	int closePSAMCard(int id)		
Description	Close the PSAM card channel and call after completing the PSAM card communication.		
Parameter	int	id	Card slot id，defined in CardReaderConstant.Psam: PSAM_CARD_1：Card channel 1 PSAM_CARD_2：Card channel 2 PSAM_CARD_3：Card channel 3
Return	int	0	Success
		Not 0	Fail
Instruction	N/A		

5.3.3.4. transmitPSAM

Function	OperationResult transmitPSAM(int id,byte[] cmd)
Description	The PSAM data transmission interface is called after the PSAM card channel is

	opened.		
Parameter	int	id	Card slot id，defined in CardReaderConstant.Psam: PSAM_CARD_1：Card channel 1 PSAM_CARD_2：Card channel 2 PSAM_CARD_3：Card channel 3
	byte[]	cmd	Transparent instruction
Return	Operation Result	0	Success，get the returned data by getData()
		Not 0	Failed
Instruction	the interface should be called after the contact card channel is opened, otherwise the status code 0x6F00 will be returned.		

5.3.4. Mifare Card interface specification

5.3.4.1. Interface list

Interface prototype	**Interface description**
verifyKeyA	verify key A
verifyKeyB	Verify key B
readBlock	Read block data
writeBlock	Write block data
writeValue	Write value
readValue	Read value
increaseValue	Increase value
decreaseValue	Decrease value
readMifareUltralightCard	Read MifareUltralight card
writeMifareUltralightCard	Write MifareUltralight card

5.3.4.2. verifyKeyA

Function	int verifyKeyA(int sectorIndex, byte[] key)		
Description	verify key A		
Parameter	int	sectorIndex	Sector index
	byte[]	key	Byte array，key data
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.3.4.3. verifyKeyB

Function	int verifyKeyB(int sectorIndex, byte[] key)		
Description	verify key B		
Parameter	int	sectorIndex	Sector index
	byte[]	key	Byte array，key data
Return	int	0	Success
		Not 0	Fail
Instruction	N/A		

5.3.4.4. readBlock

Function	OperationResult readBlock(int sectorIndex, int blockOfSector)		
Description	Read block data		
Parameter	int	sectorIndex	Sector index
	int	blockOfSector	Sector block
Return	Operation Result	0	Success，get the returned block data by getData()
		Not 0	Failed
Instruction	N/A		

5.3.4.5. writeBlock

Function	int writeBlock(int sectorIndex, int blockOfSector, byte[] buffer)		
Description	Write block data		
Parameter	int	sectorIndex	Sector index
	int	blockOfSector	Sector block
	byte[]	buffer	Data
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.3.4.6. writeValue

Function	int writeValue(int sectorIndex, int blockOfSector, byte[] money)		
Description	Write value to sector		
Parameter	int	sectorIndex	Sector index
	int	blockOfSector	Sector block
	byte[]	amount	money amount to be written
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.3.4.7. readValue

Function	OperationResult readValue(int sectorIndex, int blockOfSector)		
Description	Read sector value		
Parameter	int	sectorIndex	Sector index
	int	blockOfSector	Sector block
Return	Operation Result	0	Success, get the returned data by getData()
		Not 0	Failed
Instruction	N/A		

5.3.4.8. increaseValue

Function	int increaseValue(int sectorIndex, int blockOfSector, int value)		
Description	Increase value		
Parameter	int	sectorIndex	Sector index
	int	blockOfSector	Sector block
	int	value	Value
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.3.4.9. decreaseValue

Function	int decreaseValue(int sectorIndex, int blockOfSector, int value)		
Description	Decrease value		
Parameter	int	sectorIndex	Sector index
	int	blockOfSector	Sector block
	int	value	Value
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.3.4.10. readMifareUltralightCard

Function	OperationResult readMifareUltralightCard(int blockIndex)		
Description	Read MifareUltralight card		
Parameter	int	blockIndex	Block index
Return	Operation Result	0	Success，get the returned data by getData()
		not 0	Failed
Instruction	N/A		

5.3.4.11. writeMifareUltralightCard

Function	int writeMifareUltralightCard(int blockIndex, byte[] data)		
Description	Write MifareUltralight card		
Parameter	int	blockIndex	Block index
	byte[]	data	Data
Return	int	0	Success
		Not 0	Failed
Instruction	N/A		

5.6. EMV process operation module

5.3.5. Module introduction

The module is mainly used to proceed IC card transaction, and get IC card data information. EMV processing consists of normal mode and advanced mode. In the normal mode the card information can be obtained by directly calling the readCard interface. In the advanced mode you need to search card first, then set parameters according to the search results, and call the emvProcess interface to proceed EMV process .

5.3.6. Interface specification

5.3.6.1. Interface list

Interface prototype	**Interface introduction**
setTermConfig	Set terminal parameter
getTermConfig	Get terminal parameter
clearAID	Clear AID
clearIcAID	Clear IC AID
Clear	Clear
clearCAPK	Clear CAPK
updateAID	Update AID
updateIcAID	Update IC AID
updateRfAID	Update RF AID
updateCAPK	Update CAPK

getAID	Get AID list
getIcAID	Get IC AID list
getRfAID	Get RF AID list
getCAPK	Get CAPK list
emvProcess	EMV operation process
tradeResponse	Secondary authorization interface
emvCancel	Cancel EMV process
getEmvHashCode	Get EMV hash code
setTlv	Set TLV data to EMV kernel
getTlv	Get TLV data from EMV kernel
getTlvValue	Get TLV Value
getTlvList	Get TLV list
getTlvEncrypted	Get encrypted TLV
getTlvValueEncrypted	Get encrypted TLV value
getTlvListEncrypted	Get encrypted TLV list
queryECBalance	Query E-Cash balance
getEmvCardLog	Get EMV card log

5.3.6.2. setTermConfig

Function	boolean setTermConfig(EMVTermConfig config)		
Description	Set terminal parameters for emv transactions		
Parameter	EMVTermConfig	config	Terminal parameters, see details 6.6 EMVTermConfig
Return	boolean	true	Success
		false	Failed
Instruction	a set of default terminal parameters are stored in the terminal. If you need to modify them, please call this interface		

5.3.6.3. getTermConfig

Function	EMVTermConfig getTermConfig()
Description	get terminal parameters
Parameter	N/A

Return	EMVTerm Config	Terminal parameters, see details 6.6 EMVTermConfig
Instruction	N/A	

5.3.6.4. clearAID

Function	boolean clearAID()		
Description	Clear AID		
Parameter	N/A		
Return	boolean	true	Success
		false	Failed
Instruction	Clear parameters set by the interface updateAID		

5.3.6.5. clearIcAID

Function	boolean clearIcAID()		
Description	Clear Contact Card AID		
Parameter	N/A		
Return	boolean	true	Success
		false	Fail
Instruction	Clear parameters set by the interface updateIcAID		

5.3.6.6. clearRfAID

Function	boolean clearRfAID()		
Description	Clear contactless card AID		
Parameter	N/A		
Return	boolean	true	Success
		false	Fail
Instruction	Clear parameters set by the interface updateRfAID		

5.3.6.7. clearCAPK

Function	boolean clearCAPK()		
Description	Clear CAPK		
Parameter	N/A		
Return	boolean	true	Success
		false	Fail
Instruction	Clear parameters set by the interface updateCAPK		

5.3.6.8. updateAID

Function	boolean updateAID(byte[] AID)		
Description	Update AID		
Parameter	byte[]	AID	TLV format, see details 5.6.4.1AID/RID Parameter
Return	boolean	true	Success
		false	Failed
Instruction	The interface for updating AID by default, suitable for contact cards and contactless cards, e.g.: smartPosApi.updateAID(str2bytes("9f0608a000000333010103df0101009f08020020df1105d84000a800df1205d84004f800df130500100000009f1b0400000000df150400000000df160199df170199df14039f3704df1801019f7b06000000000000df1906000000000000df2006000999999999df2106000999999999"));		

5.3.6.9. updateIcAID

Function	boolean updateIcAID(byte[] AID)		
Description	Update Contact Card AID		
Parameter	byte[]	AID	TLV format, see details 5.6.4.1AID/RID Parameter
Return	boolean	true	Success
		false	Failed
Instruction	After calling this funtion, the contact card will preferentially use the AID set by this interface in the EMV process.		

5.3.6.10. updateRfAID

Function	boolean updateRfAID(byte[] AID)		
Description	Update Contactless Card AID		
Parameter	byte[]	AID	TLV format, see details 5.6.4.1AID/RID Parameter
Return	boolean	true	Success
		false	Failed
Instruction	After calling this funtion, the contactless card will preferentially use the AID set by this interface in the EMV process.		

5.3.6.11. updateCAPK

Function	boolean updateCAPK(byte[] CAPK)		
Description	Update CAPK		
Parameter	byte[]	CAPK	TLV format, see details 5.6.4.2CAPK Parameter
Return	boolean	true	Success
		false	Failed

Instruction	e.g.: smartPosApi.updateCAPK(str2bytes("9f0605a0000003339f22010adf05083230333 031323330df060101df070101df028180b2ab1b6e9ac55a75adfd5bbc34490e53c4c 3381f34e60e7fac21cc2b26dd34462b64a6fae2495ed1dd383b8138bea100ff9b7a1 11817e7b9869a9742b19e5c9dac56f8b8827f11b05a08eccf9e8d5e85b0f7cfa644eff 3e9b796688f38e006deb21e101c01028903a06023ac5aab8635f8e307a53ac742bdc e6a283f585f48efdf040103df0314c88be6b2417c4f941c9371ea35a377158767e4e3"));

5.3.6.12. getAID

Function	List<byte[]> getAID()	
Description	Get AID list	
Parameter	N/A	
Return	List<byte[]>	AID list
Instruction	This funtion returns the default list of AIDs.	

5.3.6.13. getIcAID

Function	List<byte[]> getIcAID()	
Description	Get IC AID list	
Parameter	N/A	
Return	List<byte[]>	AID list
Instruction	This function returns the list of AIDs set by the updateIcAID function	

5.3.6.14. getRfAID

Function	List<byte[]> getRfAID()	
Description	Get contactless AID list	
Parameter	N/A	
Return	List<byte[]>	AID list
Instruction	This function returns the list of AIDs set by the updateRfAID function	

5.3.6.15. getCAPK

Function	List<byte[]> getCAPK()	
Description	Get a list of CAPKs, including CAPK parameters set by the updateCAPK function	
Parameter	N/A	
Return	List<byte[]>	CAPK list
Instruction	N/A	

5.3.6.16. emvProcess

Function	void emvProcess(EMVTransData transData, EMVTransListener listener)		
Description	EMV process for EMV processing of contact and contactless cards		
Parameter	EMVTransData	transData	Data required by the EMV process, see details 6.7 EMVTransData
	EMVTransListener	listener	EMV process processing listener, see details 5.6.3EMVTransListener
Return	N/A		
Instruction	For an example of using this interface, see Appendix 2 Read card example		

5.3.6.17. tradeResponse

Function	int tradeResponse(String field39, String field55)		
Description	Transaction Response (Online Data Processing)		
Parameter	String	field39	response code(00)
	String	field55	issuer authentication(910abead68dd568c354d3030)
Return	int	The transaction response result, defined in EMVConstant.Status: STATUS_APPROVED：Approve STATUS_DECLINED：Decline	
Instruction	After the online transaction returns the authentication data of the issuing bank, this interface is called for authorization twice. When calling this interface, the card channel cannot be closed, and the card must be kept in place, otherwise it will fail.		

5.3.6.18. emvCancel

Function	void emvCancel()
Description	Cancel the emv process
Parameter	N/A
Return	N/A
Instruction	Not yet implemented

5.3.6.19. getEmvHashCode

Function	int getEmvHashCode()	
Description	Get emv hash code	
Parameter	N/A	
Return	int	emv hash code
Instruction	Not yet implemented	

5.3.6.20. setTlv

Function	int setTlv(byte[] tag, byte[] value)		
Description	Set TLV to EMV kernel		
Parameter	byte[]	tag	The tag to be set
	byte[]	value	value
Return	int	0	Success
		Non-0	Failed
Instruction	N/A		

5.3.6.21. getTlv

Function	byte[] getTlv(int tag)		
Description	Get TLV data from EMV kernel		
Parameter	int	tag	The tag value to be read
Return	byte[]	Tag and corresponding value, TLV format	
Instruction	N/A		

5.3.6.22. getTlvValue

Function	byte[] getTlvValue(int tag)		
Description	Get the value corresponding to the tag from the EMV kernel		
Parameter	int	tag	The tag to be read
Return	byte[]	The value corresponding to the tag, the tag and length are removed, and only the value is retained	
Instruction	N/A		

5.3.6.23. getTlvList

Function	byte[] getTlvList(int[] tags)		
Description	Get TLV list data		
Parameter	int[]	tags	The set of tags to be read
Return	byte[]	TLV data set corresponding to tags	
Instruction	N/A		

5.3.6.24. getTlvEncrypted

Function	String getTlvEncrypted(int tag, int keyId, int calMode, byte[]... vector)		
Description	Get encrypted TLV data		
Parameter	int	tag	The tag to be acquired
	int	keyId	The specified encryption key id, defined in

			PinPadConstant.KeyId: KEY_ID_FIRST_MAINKEY KEY_ID_SECOND_MAINKEY KEY_ID_FIRST_TDKEY KEY_ID_FIRST_PINKEY KEY_ID_FIRST_MACKEY DUKPT key id, defined in PinPadConstant.Dukpt: DUKPT_PIN_KEY DUKPT_MAC_KEY_REQ DUKPT_MAC_KEY_RES DUKPT_DAT_KEY_REQ DUKPT_DAT_KEY_RES
	int	calMode	Encryption mode, defined in PinPadConstant.EncryptMode: ENCRYPT DECRYPT DUKPT_ENCRYPT DUKPT_DECRYPT
	byte[]	vector	Vector, which needs to be passed in for CBC encryption
Return	byte[]	Encrypted TLV data	
Instruction	N/A		

5.3.6.25. getTlvValueEncrypted

Function	String getTlvValueEncrypted(int tag, int keyId, int calMode, byte[]... vector)		
Description	Get encrypted TLV value data		
Parameter	int	tag	The tag to be acquired
	int	keyId	The specified encryption key id, defined in PinPadConstant.KeyId: KEY_ID_FIRST_MAINKEY KEY_ID_SECOND_MAINKEY KEY_ID_FIRST_TDKEY KEY_ID_FIRST_PINKEY KEY_ID_FIRST_MACKEY DUKPT key id, defined in PinPadConstant.Dukpt: DUKPT_PIN_KEY DUKPT_MAC_KEY_REQ DUKPT_MAC_KEY_RES DUKPT_DAT_KEY_REQ DUKPT_DAT_KEY_RES
	int	calMode	Encryption mode, defined in PinPadConstant.EncryptMode: ENCRYPT

			DECRYPT DUKPT_ENCRYPT DUKPT_DECRYPT
	byte[]	vector	Vector, which needs to be passed in for CBC encryption
Return	byte[]	Encrypted TLV value data	
Instruction	N/A		

5.3.6.26. getTlvListEncrypted

Function	String getTlvListEncrypted(int[] tags, int keyId, int calMode, byte[]... vector)		
Description	Get encrypted TLV list data		
Parameter	int[]	tags	A collection of tags to be acquired
	int	keyId	The specified encryption key id, defined in PinPadConstant.KeyId: KEY_ID_FIRST_MAINKEY KEY_ID_SECOND_MAINKEY KEY_ID_FIRST_TDKEY KEY_ID_FIRST_PINKEY KEY_ID_FIRST_MACKEY DUKPT key id, defined in PinPadConstant.Dukpt: DUKPT_PIN_KEY DUKPT_MAC_KEY_REQ DUKPT_MAC_KEY_RES DUKPT_DAT_KEY_REQ DUKPT_DAT_KEY_RES
	int	calMode	Encryption mode, defined in PinPadConstant.EncryptMode: ENCRYPT DECRYPT DUKPT_ENCRYPT DUKPT_DECRYPT
	byte[]	vector	Vector, which needs to be passed in for CBC encryption
Return	byte[]	Encrypted TLV list data	
Instruction	N/A		

5.3.6.27. queryECBalance

Function	Balance queryECBalance(int channelType)		
Description	Check electronic cash balance		
Parameter	int	channelType	Channel type, defined in EMVConstant.ChannelType: CONTACT CONTACTLESS

Return	Balance	Electronic cash balance information, see details 6.11 Balance
Instruction	N/A	

5.3.6.28. getEmvCardLog

Function	List<EMVCardLog> getEmvCardLog(int channelType)		
Description	Get EMV card log		
Parameter	int	channelType	Channel type, defined in EMVConstant.ChannelType: CONTACT CONTACTLESS
Return	List<EMVCardLog>	EMV card log, see details 6.5 EMVCardLog	
Instruction	N/A		

5.3.7. EMVTransListener

5.3.7.1. Interface list

Interface prototype	**Interface introduction**
onWaitAppSelect	Waiting for app selection
onInputAmount	Waiting to input the amount
onConfirmEC	Confirm the use of electronic cash
onConfirmCardNo	Confirm card number
onCertVerfiy	verify identity
onCardHolderPwd	Enter password
onEmvTerGetInputDisp	Get display data
onOffLinePinVerify	Offline PIN Verification Results
onOffLinePinLastTimeVerify	Offline PIN Remaining Verification Times
onTransResult	Process result

5.3.7.2. onWaitAppSelect

Function	int onWaitAppSelect(List<String> appNameList, boolean isFirstSelect)		
Description	Wait for the application selection, EMV judges the return value, and enters the next step of the process.		
Parameter	List<String>	appNameList	EMV kernel returns application list
	boolean	isFirstSelect	Whether it is the first time to choose the

			application
Return	int	>=0	Selected app index
		-1	Cancel
		-2	Failed
Instruction	N/A		

5.3.7.3. onInputAmount

Function	long onInputAmount()	
Description	Waiting for the input amount, when calling back this interface, you need to return the input amount.	
Parameter	N/A	
Return	long	Amount, minimum unit of currency
Instruction	N/A	

5.3.7.4. onConfirmEC

Function	int onConfirmEC()		
Description	Confirm using electronic cash. The return value is set by the application. EMV judges the return value, and enter the next step of the process.		
Parameter	N/A		
Return	int	0	Confirm
		-1	Cancel
Instruction	N/A		

5.3.7.5. onConfirmCardNo

Function	int onConfirmCardNo(String cardNo)		
Description	Confirm the card number. The return value is set by the application. EMV judges the return value, and enter the next step of the process.		
Parameter	String	cardNo	Card number to be confirmed
Return	int	0	Confirm
		-1	Cancel
Instruction	N/A		

5.3.7.6. onCertVerfiy

Function	int onCertVerfiy(String certType, String certValue)		
Description	Identity authentication. The return value is set by the application. EMV judges the return value, and enter the next step of the process.		
Parameter	String	certType	Type of certificate
	String	certValue	Content of certificate

Return	int	0	Confirm
		-1	Cancel
Instruction	N/A		

5.3.7.7. onCardHolderPwd

Function	PINResult onCardHolderPwd(boolean isOnlinePin, int offlinePinLeftTimes)		
Description	If it is online, enter the password, and the application will set the result and calculate the PINBLOCK to return, and EMV will judge the return value and enter the next step of the process. If it is offline, then EMV internally proceed,without judging the return value.		
Parameter	boolean	isOnlinePin	Whether it is an online transaction
	int	offlinePinLeftTimes	Offline Password Remaining Times
Return	PINResult	See details 6.9 PINResult	
Instruction	N/A		

5.3.7.8. onEmvTerGetInputDisp

Function	void onEmvTerGetInputDisp(byte[] bytes)		
Description	Get display data		
Parameter	byte[]	bytes	Data to be displayed
Return	N/A		
Instruction	N/A		

5.3.7.9. onOffLinePinVerify

Function	void onOffLinePinVerify(boolean isPinOK)		
Description	Offline password verification		
Parameter	boolean	isPinOK	Verification results
Return	N/A		
Instruction	verify the password through the interface when the password is needed		

5.3.7.10. onOffLinePinLastTimeVerify

Function	void onOffLinePinLastTimeVerify(int number)		
Description	Offline password remaining input times		
Parameter	int	number	remaining times
Return	N/A		
Instruction	If the number of verifications is exceeded and the password is not entered correctly, the card will be locked.		

5.3.7.11. onTransResult

Function	void onTransResult(int status, TransResult transResult)		
Description	Emv processing result		
Parameter	int	status	Defined in EMVConstant.Status: STATUS_APPROVED STATUS_DECLINED STATUS_ONLINE_REQUEST STATUS_END_APPLICATION STATUS_SELECT_NEXT STATUS_TRY_ANOTHER_INTERFACE STATUS_TRY_AGAIN STATUS_NA STATUS_TC_TO_ARQC STATUS_TC_TO_AAC STATUS_ARQC_ODA
	TransResult	transResult	See details 6.12 TransResult
Return	N/A		
Instruction	N/A		

5.3.8. Parameter Description

5.3.8.1. AID/RID Parameter

The AID / RID parameters include the tags below. When using the Master Kernel, in addition to the TAGs defined in the table below, it also supports 3Byte length TAGs starting with DF or FF as defined by the Master specification.

TAG	NAME	FORMAT	LENGTH (Byte)
9F06	AID (Application Identifier)	B	5-16
DF01	ASI (Application Selection identifier)	B	1
9F09	Application Version Number	B	2
DF11	TAC-Default	B	5
DF12	TAC -Online	B	5
DF13	TAC-Denial	B	5
9F1B	Terminal Floor Limit	B	4
DF15	Threshold Value for Biased Random Selection	B	4
DF16	Biased Random Selection Maximum Target Percentage	CN	1
DF17	Biased Random Selection Target Percentage	CN	1
DF14	Default DDOL	B	Var..252
9F7B	E-Cash Balance Transaction Limit	CN	6

DF19	Reader Contactless Floor Limit	CN	6
DF20	Reader Contactless Transaction Limit	CN	6
DF21	Reader CVM Required Limit	CN	6

5.3.8.2. CAPK Parameter

CAPK parameters can include the following TAGs.

TAG	NAME	FORMAT	LENGTH (Byte)
9F06	RID (Registered Application Provider Identifier)	B	5
9F22	CA Public Key Index	B	1
DF05	CA Public Key Expiration Date	N8	4
DF06	CA Public Key Hash algorithm identifier	B	1
DF07	CA Public Key Signature algorithm identifier	B	1
DF02	CA Public Key	B	Var..248
DF04	CA Public Key Exponent	B	1/3
DF03	CA Public Key Check Value	B	Var

5.7. Scanner module

5.3.9. Module introduction

Scanner module mainly scans bar code or QR code data provided by customers, analyzes and returns results.

5.3.10. Interface specification

5.3.10.1. Interface list

Interface prototype	**Interface specification**
startScan	Start scanning the code
stopScan	Stop scanning the code

5.3.10.2. startScan

Function	void startScan(Context context, Bundle bundle,ScanResultListener listener)		
Description	Start scanning the code, mainly used to obtain the payment code during transaction process.		
Parameter	Context	context	Context
	Bundle	bundle	Set configuration as needed, used as follows: ScannerConfig scannerConfig = ScannerConfig.create()

			.setScanCameraId(ScannerConstant.CameraId.BACK) .isFlashOn(true) .setTimeout(60) .isShowCamaraIcon(true);
	ScanResultListener	listener	Scancode result listener, see details 5.7.2 ScanResultListener
Return	N/A		
Instruction	N/A		

5.3.10.3. stopScan

Function	void stopScan()
Description	Stop scanning the code
Parameter	N/A
Return	N/A
Instruction	N/A

5.3.11. ScanResultListener

5.3.11.1. Interface list

Interface prototype	**Interface specification**
onGetScanResult	Scancode result callback

5.3.11.2. onGetScanResult

Function	void onGetScanResult(ScanResult scanResult)		
Description	The code scanning interface is called back, and the user obtains the relevant results of the code scanning.		
Parameter	ScanResult	scanResult	scanResult：Including resultCode and qrContent, where: resultCode：Scan code result, defined in ScannerConstant.ResultCode: SUCCESS CANCEL TIMEOUT ERROR qrContent：QR code data
Return	N/A		

Instruction	N/A

5.4. Led module

5.4.1. Module introduction

This module provides the switches of the lights of red, green, yellow and blue in the upper right corner of the device.

5.4.2. Interface specification

5.4.2.1. Interface list

Interface prototype	**Interface specification**
setLed	Set led

5.4.2.2. setLed

Function	boolean setLed(int light, boolean isOn)		
Description	Set the LED Switch		
Parameter	int	light	Defined in LedLightConstant: RED：Red light operating bit GREEN：Green light operating bit YELLOW：Yellow light operating bit BLUE：Blue light operating bit
	boolean	isOn	true, open false, close
Return	boolean	true	Success
		false	Failed
Instruction	The interface currently only supports red and blue LED		

5.5. Beeper module

5.5.1. Module introduction

The module is used to call the device beeper, whch can issue a beeper hint in transaction.

5.5.2. Interface specification

5.5.2.1. Interface list

Interface prototype	**Interface specification**
beep	Beep
release	Release beeper

5.5.2.2. beep

Function	void beep()
Description	Beep once
Parameter	N/A
Return	N/A
Instruction	N/A

5.5.2.3. beep

Function	void beep(int loop, float rate , float voice)		
Description	Set the number of loops, play speed and volume for the beep.		
Parameter	int	loop	Number of cycles
	float	rate	Play speed (0.5-2.0)
	float	voice	volume(0.0-1.0)
Return	N/A		
Instruction	N/A		

5.5.2.4. beep

Function	void beep(int sourceId ,int loop, float rate , float voice)		
Description	Set the number of loops, play speed and volume for passed-in audio sources.		
Parameter	int	sourceId	Audio source id
	int	loop	Number of loops
	float	rate	Play speed (0.5-2.0)
	float	voice	volume(0.0-1.0)
Return	N/A		
Instruction	N/A		

5.5.2.5. releaseBeeper

Function	void releaseBeeper()

Description	Release beeper
Parameter	N/A
Return	N/A
Instruction	Call this interface when you no longer need to play the sound.

5.6. Tag module

5.6.1. Module introduction

This module is used for reading, writing and verification of TAG.

5.6.2. Interface specification

5.6.2.1. Interface list

Interface prototype	**Interface specification**
writeTagData	Write Tag Data
readTagData	Read Tag Data
verifyTag	Verify the legitimacy of the TAG

5.6.2.2. writeTagData

Function	boolean writeTagData(int address , byte[] data)		
Description	Write Tag Data		
Parameter	int	address	The offset of the data to be written
	byte[]	data	Data to be written
Return	boolean	true	Success
		false	Failed
Instruction	N/A		

5.6.2.3. readTagData

Function	OperationResult readTagData(int address, int len)		
Description	Read Tag Data		
Parameter	int	address	The offset of the data to be read
	int	len	The length of the data to be read
Return	Operation Result	0	Success，get the tag data by getData()
		Not 0	Failed
Instruction	N/A		

5.6.2.4. verifyTag

Function	boolean verifyTag()		
Description	Verify the legitimacy of the TAG		
Parameter	N/A		
Return	boolean	true	Success
		false	Failed
Instruction	N/A		

6. Entity class

6.1. Introduction

Description and explanation of entity class that the user encountered when caling the SDK interfaces.

6.2. Class list

Class name	**Function introduction**
OperationResult	Operation result
CardInfo	Card info
EMVCardLog	EMV card log
EMVTermConfig	EMV terminal configuration
EMVTransData	EMV transaction data
PINResult	EMV password data
PinPadResult	Pinpad result callback
Balance	E-cash balance
DeviceInfo	Device Information
DeviceVersion	Device version information
PrintElement	element to print
ScanResult	Scancode result

6.3. OperationResult

Member variable	**Type**	**Description**
statusCode	int	Status code

data	String	Data

6.4. CardInfo

Member variable	Type	Description
cardType	String	Card type (magnetic strip card /IC card)
cardNo	String	Card number
cardValidThru	String	Card expiration date
cardHolderName	String	Cardholder's Name
encTrack1	String	Track one
encTrack2	String	Track two
encTrack3	String	Track three
icData55	String	IC card 55 field
cardSequence	String	Card serial number
responseCode	String	Card reader response code
comments	String	Remarks
serviceCode	String	Service code

6.5. EMVCardLog

Member variable	Type	Description
isAmountExist	boolean	mark of Transaction amount exists
amount	long	Transaction amount
isOtherAmountExist	boolean	mark of other amount exists
otherAmount	long	Other amount
isDateExist	boolean	mark of transaction date（YYMMDD）exists
transDate	String	Transaction date（YYMMDD）
isTimeExist	boolean	mark of transaction time exists
transTime	String	Transaction time
isCountryCodeExist	boolean	mark of country code exists
countryCode	String	Country code（9F1A）
isCurrencyExist	boolean	mark of currency code exists
currencyCode	String	Currency code（5F2A）
isATCExist	boolean	mark of transaction counter exists
ATC	String	Transaction counter（9F36）
isTransTypeExist	boolean	mark of transaction type exists
transType	byte	Transaction type（9C）
isMerchNameExist	boolean	mark of merchant name exists
merchName	String	Merchant name（9F4E）
otherTLVLog	String	Other data elements not defined in this structure are stored in the TLV according to the format of the TLV list

6.6. EMVTermConfig

Member variable	Type	Description
referCurrCon	long	Reference currency code and transaction code conversion coefficient (trading currency to reference the currencies of * 1000)
merchName	String	Merchant name
merchCateCode	String	Merchant category code (no requirement is not set)
merchId	String	Merchant id，9F16
termId	String	Terminal id，9F1C
termType	int	Terminal type, indicating terminal environment, communication capability and operational control，9F35
capability	String	Terminal Capability, indicating card data input, CVM and security capabilities of the terminal，9F33
extCapability	String	Additional terminal capability, indicating terminal data input and output capabilities，9F40
transCurrCode	String	Transaction currency code, indicating the transaction currency code according to ISO 4217，5F2A
transCurrExp	int	Transaction currency exponent representing the implied decimal point position from the right of the transaction amount as specified in ISO 4217，5F36
referCurrCode	String	Reference currency code
referCurrExp	int	Reference currency exponent
countryCode	String	Terminal country code, indicating the terminal country code according to ISO3166，9F1A
extField	Map<String , Object>	Extended field

6.7. EMVTransData

Member variable	Type	Description
amount	long	Transaction amount
otherAmount	long	Other amount
transType	byte	Transaction type
transDate	String	Transaction date，YYYYMMDD
transTime	String	Transaction time，HHMMSS
traceNo	String	Transaction serial number
isSupportSM	boolean	N/A
isCardAuth	boolean	Whether to perform offline data authentication
isForceOnline	boolean	Whether to forced network connection
isSupportEC	boolean	Whether to use e-cash
isSupportCVM	boolean	whether to execute CVM

flow	int	Flow type, defined in EMVConstant.Flow
channelType	int	Channel type, defined in EMVConstant.ChannelType
extField	Map<String, Object>	Extended field

6.8. OnlineResult

Member variable	Type	Description
resultCode	int	Online transaction result 0: success -1: cancel -2: fail
field55	byte[]	55 field data
Field39	byte[]	39 field data

6.9. PINResult

Member variable	Type	Description
resultCode	int	Result code
pindata	byte[]	Pinblock

6.10. PinPadResult

Member variable	Type	Description
resultCode	int	Result code
pinBlock	String	Pinblock

6.11. Balance

Member variable	Type	Description
firstCurrencyCode	String	First currency code
firstCurrencyBalance	long	First currency balance
secondCurrencyCode	String	Second currency code
secondCurrencyBalance	long	Second currency balance

6.12. TransResult

Member variable	Type	Description
kernelType	int	Kernel type, defined in EMVConstant.KernelType: KERNEL_TYPE_NULL: no matching kernel KERNEL_TYPE_EMV: EMV kernel KERNEL_TYPE_QUICS: QUICS kernel

		KERNEL_TYPE_VISA：VISA kernel KERNEL_TYPE_MASTER：MASTER kernel
resultCode	int	The result, defined in EMVConstant.ResultCode: ONLINE_REQUEST：online request OK：success ERR：general error ERR_PARAMETER：parameter error ERR_OUT_OF_MEMORY：out of memory ERR_TIMEOUT：timeout ERR_NONE_DATA：no data ERR_PROCESSING：emv process is in progress ERR_KERNEL_RESTART：kernel restart
errL1	int	errL1
errL2	int	errL2
errL3	int	errL3
sw1Sw2	int	sw1Sw2
cvm	int	CVM，defined in EMVConstant.Cvm： CVM_NO_CVM：no need CVM CVM_SIGNATURE：sign CVM_ONLINE_PIN：online PIN CVM_CONFIRM_CODE_VERIFIED：Verify confirmation code CVM_NA：CVM：N/A
cmd	int	cmd
identify	int	identify

6.13. DeviceInfo

Member variable	Type	Description
vendor	String	Manufacturer name
model	String	Terminal model
androidSN	String	Android SN
imei	String	IMEI
pluginInformation	String	Resolution
currentLocation	String	Current location

6.14. DeviceVersion

Member variable	Type	Description
androidVersion	String	Android version
customVersion	String	System customized version
posServiceVersion	String	PosService version
BootVerion	String	Boot version

SoftVersion	String	Main app version
SubVersion	String	Sub app version

6.15. ScanResult

Member variable	Type	Description
resultCode	int	Scan code result
qrContent	String	QR code data

6.16. PrintElement

Member variable	Type	Description
text	String	string to print
bitmap	Bitmap	Printed image content
textSize	int	Print font size Defined in PrinterConstant.FontSize: FONT_SIZE_SMALL FONT_SIZE_MIDDLE FONT_SIZE_LARGE FONT_SIZE_HUGE FONT_SIZE_SMALL_HEIGHT FONT_SIZE_SMALL_WIDTH FONT_SIZE_MIDDLE_HEIGHT FONT_SIZE_MIDDLE_WIDTH FONT_SIZE_LARGE_HEIGHT FONT_SIZE_LARGE_WIDTH FONT_SIZE_HUGE_HEIGHT FONT_SIZE_HUGE_WIDTH
align	int	Alignment (default left) Defined in PrinterConstant.Align: ALIGN_LEFT ALIGN_CENTER ALIGN_RIGHT
isBold	boolean	bold switch
isItalic	boolean	italic switch
isUnderline	boolean	underline switch
isStrikeout	boolean	strikeout switch

Constructor	public PrintElement(String text);
	public PrintElement(String text, int align);
	public PrintElement(String text, int align, int textSize);
	public PrintElement(String text, int align, int textSize, boolean isBold);
	public PrintElement(Bitmap bitmap);	

	public PrintElement(Bitmap bitmap, int align);

6.17. KeyInfo

Member variable	Type	Description
x	int	The x coordinate of the upper left corner of the key
y	int	The y coordinate of the upper left corner of the key
w	int	Key width
h	int	key height
keyFuc	String	Key Function
keyText	String	key text
txvSize	int	key text size
txvColor	int	key text color
bgColor	int	key background color
strokeColor	int	key frame color
strokeWidth	int	key frame width
Constructor	public KeyInfo(int x, int y, int w, int h, String keyFuc); public KeyInfo(int x, int y, int w, int h, String keyFuc, int txvSize, int txvColor, int bgColor, int strokeColor, int strokeWidth);	

7. status code

7.1. Card reading status code(cardInfo.getResponseStatus)

CardReaderConstant.ReadCardRes	Status code	Status Description
READ_CARD_SUCCESS	00	Card read successfully
CARD_TIME_OUT	01	Card read timeout
CARD_CANCEL	02	Card read timeout
CARD_FAILED	03	Card read fail
CARD_MAGNETIC_FAILED	04	Failed to read magnetic stripe card information
CARD_PBOC_FAILED	05	EMV process failed
AMOUNT_LENGTH_ERROR	06	Amount length error
TRADE_TIME_ERROR	07	Transaction time error
OPEN_CARD_READER_FAIL	08	Open card reader channel fail
SDK_NEED_UPDATE	09	SDK need to be updated
READ_CARD_EXCEPTION	10	Card reader interface exception

7.2. Pin pad status code(pinPadResult.getResultCode)

PinPadConstant.PinPad	Status code	Status Description
SUCCESS	0	Success
BYPASS	1	Confirm without password
CANCEL	-1	Cancel inputting password
TIMEOUT	-2	Input password timeout
ERROR	-3	Error

7.3. Firmware upgrade status code(FirmwareUpgradeConstant)

FirmwareUpgradeConstant	Status code	Status Description
VERSION_NOT_MATCH	01	Version numbers do not match
CHILD_APP_NO_NEED_UPDATE	02	No need to upgrade sub-app
GET_DEVICE_VERSION_FAIL	03	Failed to get device version
COMMUNICATION_FAIL	04	Communication failed
EXCEPTION	05	Program exception
GET_MAIN_APP_VERSION_FAIL	06	The download of the main application succeeded, but the acquisition of the version number of the main application failed
UPDATE_MAIN_APP_FAIL	07	The parameters of the backup area are downloaded successfully, but the update of the main application failed
GET_CHILD_ID_COMMUNICATION_FAIL	08	The update of the sub-app is successful, but the communication of obtaining the sub-app ID failed
UPDATE_DEVICE_FAIL	09	Download sub-app parameters succeeded, device update failed
GET_CHILD_ID_FAIL	10	The update of the sub-app succeeds, but the acquisition of the sub-app ID fails
FILE_FORMAT_ERROR	11	Incorrect file format
GET_PN_FAIL	12	Failed to get pn
GET_FILE_PN_FAIL	13	Failed to get pn in file
GET_SIGN_FLAG_FAIL	14	Failed to get signature ID
DOWNLOAD_BACKUP_AREA_FAIL	15	Failed to download backup area

GET_XOR32_FAIL	16	Failed to download backup area to get applen or xor32 or encryptTransKey
SIGN_DATA_LENGTH_ERROR	17	signData length error

7.4. Operation result status code(operationResult.getStatusCode)

ResultCode	Status code	Status Description
SUCCESS	0	Success
FAIL	1	Failed
PARAMS_ERROR	2	Parameter error

Appendix 1 Print Instruction

1. Initialization

The printer initialization calls the initPrinter() interface, which needs to be called back successfully before printing.

Sample code:

```
smartPosApi = ITYSmartPosApi.get(getApplicationContext());
smartPosApi.initPrinter(new MyPrinterInitListener());
```

2. Print example (recommended)

```
smartPosApi.setLineSpace(10);
smartPosApi.appendPrintElement(new PrintElement("Hello!"));
smartPosApi.appendPrintElement(new PrintElement("This is my printer!", PrinterConstant.Align.ALIGN_CENTER));
smartPosApi.appendPrintElement(new PrintElement(bitmap));
int status = smartPosApi.startPrintElement();
```

3. Print script format specification

3.1 Command overview

Prefix	Command	Parameter 1	Parameter 2	Parameter 3	Parameter 4	Description
!	hz	The font size (s, n, l, sn, sl, nl) S (small font) : a line of print 24; N (standard font) : a line of print 16; L (large font) : a line of print 12; Sn: small font width, standard font height; Sl: small font width, large font height; Nl: standard font width, large font height;				Chinese style
	asc	The font size (s, n, l, sn, sl, nl) S (small font) : a line of print 48;				Western style

		N (standard font) : a line of print 32; L (large font) : a line of print 24; Sn: small font width, standard font height; Sl: small font width, large font height; Nl: standard font width, large font height;				
	gray	[1, 10] The default 5				Gray/concentration and thickness (support for only the bold printer for the following operations: 1 ~ 5, 6 ~ 10 is bold)
	yspace	[0, 60] The default 6				Line spacing
	barcode	A single barcode width (from the point of effect, suggest the width > = 2) [1] The default 2	Expect highly [1320] and must be in multiples of 8 The default is 64			Barcode style

	qrcode	Expect highly (int) No more than print borders The default is 100	Error correction level [0, 3] General is 2			Qr code style
	NLFONT	Chinese characters font (corresponding to the font enumeration values)	English font (corresponding to the font enumeration values)	Print mode (optional) 0 - in longitudinal and transverse magnification zoom in 1 - in longitudinal and transverse magnification is normal 2, transverse, longitudinal normal amplification 3, transverse, longitudinal normal to normal		
	NLPRNOVER					Printing the paper feed, print the data out of the paper

*	text	Display position (r, l, c, X: offset) L: left C: in the middle R: right "X: offset" mode, the offset for a specific value, 0 ~ 384, for example: x: 120	The character data (string)			Print the word
	underline	Display position (l,c,r, X: offset) L: left C: in the middle R: right "X: offset" mode, the offset for a specific value, 0 ~ 384, for example: x: 120	The character Data (string)			Underlined text printing
	feedline	The number of rows (int)				Into/out of paper (according to the number of rows) Positive for feed, negative out of paper
	line					Print out a delimiter

	barcode	Display position (r, l, c, X: offset) L: left C: in the middle R: right "X: offset" mode, the offset for a specific value, 0 ~ 384, for example: x: 120	Data (String)			Print the barcode
	qrcode	Display position (r, l, c, X: offset) L: left C: in the middle R: right "X: offset" mode, the offset for a specific value, 0 ~ 384, for example: x: 120	Data (String)			Print the barcode
	image	Display position (r, l, c, X: offset) L: left C: in the middle	The size of the (X * Y) X: width Y: highly	Image data (# logo: data:, path:) # logo: look for equipment marked logo image data.		Print the image

		R: right "X: offset" mode, the offset for a specific value, 0 ~ 384, for example: x: 120	Such as: 128 * 64	Data: based on the BMP monochrome image data format, the format is as follows: Data: data encoding;The data. Among them: Data coding is as follows: base64 The sample is as follows: Data: base64.R0lGODlhuA ...; Path: followed by to print PNG images in the android file system on an absolute address, such as: path: z: 95;/ data/share/ys. PNG. If yz: domain manually threshold, yz: ' 'back to need		

				to set the threshold, and with ";'end, behind to keep up with the image file path, not' yz: 'domain automatically set threshold.		
	pause	The pause time (seconds)				Suspend specified time (applicable scenarios: multigang print, hit merchants league first, call user group)

3.2 Instructions

1) Command classification

"!"means command set type, used to set the style. Once set, the subsequent printing will be affected.

"*" means output type command, which is used to output text, images, bar code or qr code.

2) Command symbol

It and classification prefix constitute the only Identification. According to this symbol, the terminal calls the corresponding driver interface.

3) Parameter

Each print command needs passed-in parameters, and the number of parameters is fixed. The number of parameters in different commands ranges from 1 to 4. There is one space between parameters.

3.3 Output

There is a newline character for any command output. If the content is used to represent multiple lines, it needs to escape newline symbol, use "\ n" as newline characters.

3.4 Sample script

```
smartPosApi.appendScript("!hz n\n" +
       "!asc n\n" +
```

```
        "!gray 8\n" +
        "!yspace 6\n" +
        "!NLFONT hz 0\n" +
        "!NLFONT asc 0\n" +
        "*image                                                c                                                    284*81
data:base64;iVBORw0KGgoAAAANSUhEUgAAAX8AAABtCAMAAAB+85FxAAAAGXRFWHRTb2Z0d2FyZQBBZG9iZSBJbWFnZVJlYWR5ccllPAAAAAZQTFRFAAA
A////pdmf3QAABbBJREFUeNrsncl23TgMRFH//9O96O7ElgigMIiU/aBNjofY0i2wiIFKBHOdvGQQDP/h
P9fwH/5zDf/hP9fwH/5z7eUvMlIc5j8SHOc/GpznPxIc5z8a7M9/REaCl/EfDc7zHwmO8x8NttS/IiPBm/mPAsP/N/PH8H85/xFg+A//uR7iPxvA8B/+w18
hRWKI0RJEBHgxFfv+la/9+ez3H1bhL0/yl4/gjyb+DMDhr/CX0JUmKPjRBnS5sTR/udnP8G/mD1OWJv7yqfz/vTs1HK/2YvFHnn+8gpXa9pHcf9r2GY8/Av
ylnf/qFmv8pcpf94KUAln+WEihEVp8XjMT91HxOH8ytnsEkHr8/5VCI0TzXz2E90yCbgNC0wKI/ypn//1LTb4EPWiT1J+w0L7s5S+bHYjO/pSvIvLtDn8n1
VUeSILh6+HYKoALxuWPSJJA2bx+H+jgLxv5S9T+n4n/q85eGaHXGhv48wLU+es4zfw/HP9KZorb03ipEMUfm/hjB38m/9HyGVmXBgvtqLW5iT8f2tUFID+I
P9r4C8F/iwF53TSy/+DzV6tkM6/Mx396AUStpYE/2zQw6t8If2H5S8H/i/y3GVCgaQM7/sXm727Y4X6RzR9l/htrsDj/bP1l8pdt/MV15G382fwfXfzdMUI
Pf5T5n9mBIewEN8WfqL/4wdsb+ONn8GeGb4Lb+qOqAfpWia08Htq1BZDlDzWirdmmO/xk/5qb/3TwP2FAy6ybyf89kOJsxW3nf3bz7+tBqFVPiL9Suyl3Fp
0oLUVODFGok197ehBeawxK/g9jR309f+6A2N4U1EZm119em5/iL7384wtAfit/rVv0Yv77pgAaf7CJu94rk6WJafyb/ScugBxZACp/7OZ/T0GobDV1joHqd
29qwtH83b4Q7PmXz5/Cv4X/9hLAGjda7Uytt1zjjw7+MQHS1vJC/pLjLwf550O7pwfBf/QVnTVnxPq7df7s9kjzjwiQD+3mHpA6jlzdmT7mCuf/N8LV/GcX
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```

```
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\n" +
"Merchant name: * text l day test yu merchants \n" +
"* text l merchant code: 123456789012345 \n" +
"* text l terminal number: 12345678 \n" +
"* text l flight number: 201707311234 \n" +
"* text l issuer: China merchants bank line: bank of China, \n" +
"* text l validity: 20190731 \n" +
"* text l card number: China merchants bank \n" +
"* text l 6225 75 * * * * * * 5838 / C \n" +
"* text type l: consumer \n" +
"* text l certificate no. : 000001 authorization code: 415745 \n" +
"* text l batch no. : 000045 reference number: 458612356325 \n" +
"* text l trade date: 2014/07/31 14:22:02 \n" +
"!Hz n \n"+
"!Asc n \n"+
"!gray 8 \n " +
"* text l amount: \n" +
```

```
"* text l RMB: 0.01 \n" +
"* text l case note: \n" +
"* text c...\n"+
"* text l cardholder signature:");
smartPosApi.startPrint();
```

Appendix 2 Read card example

The SDK provides two ways to implement the card reading functions. The first one directly calls the card reading interface encapsulated in the SDK and obtains the card data. The second way is for users to customize the card reading process according to the relevant interface provided by SDK to obtain the card data and complete the card reading function.

4. Call the SDK card reader interface

① Open the card reader channel

```
smartPosApi.setReadCardConfig(true, true, true);
```

②Get card information

```
CardInfo cardInfo = smartPosApi.readCard("1", "20171026101010", (byte) 0, (byte) 60);
```

③IC card secondary authorization

```
smartPosApi.tradeResponse("field39", "field55");
```

④ Close the card channel after reading card

```
smartPosApi.cancelReadCard();
```

5. User customized card reading

2.1 Magnetic stripe cards

① Open the card reader channel to start card search:

```
smartPosApi.startSearchCard(true, true, true, 60);
```

② Get card information:

```
CardInfo cardInfo = smartPosApi.getMagCardData();
```

③ Close the card channel after reading card:

```
smartPosApi.stopSearchCard();
```

2.2 IC Card

①Open the card reader channel to start card search:

```
smartPosApi.startSearchCard(true, true, true, 60);
```

②Call emv interface:

```
emvTransData.setAmount(1); //Set transaction amount
emvTransData.setTransType((byte) 0x00); //The first byte of the transaction processing code in domain 3.
emvTransData.setTransDate("20201230"); //Transaction date, yyyyMMdd
emvTransData.setTransTime("103211"); //Transaction time, hhmmss
emvTransData.setSupportSM(false);//Support national key or not
emvTransData.setCardAuth(false); //Execute offline data authentication or not
emvTransData.setForceOnline(true); //Force online or not
emvTransData.setSupportEC(false); //Electronic cash transaction or not
emvTransData.setSupportCVM(true);//Execute CVM or not
emvTransData.setFlow(EMVConstant.STANDARD_EMV); //Flow type, 0x01: standard authorization process; 0x02: simple process;
0x03: q process
emvTransData.setChannelType(EMVConstant.CONTACT);   //Channel type, 0:  contact type ; 1:  non-contact type
smartPosApi.startSearchCard(true, true, true, 60);
smartPosApi.emvProcess(emvTransData, new MyEmvListener()); //Set emv callback listener
```

③Open the card reader channel to start card search:

```
class MyEmvListener implements EMVTransListener {

    @Override
    public int onWaitAppSelect(List<String> appNameList, boolean isFirstSelect) {
        return 1;
    }

    @Override
    public long onInputAmount() {
        return 1;
    }

    @Override
    public int onConfirmEC() {
        return 0;
    }

    @Override
    public int onConfirmCardNo(String cardNO) {
        return 0;
```

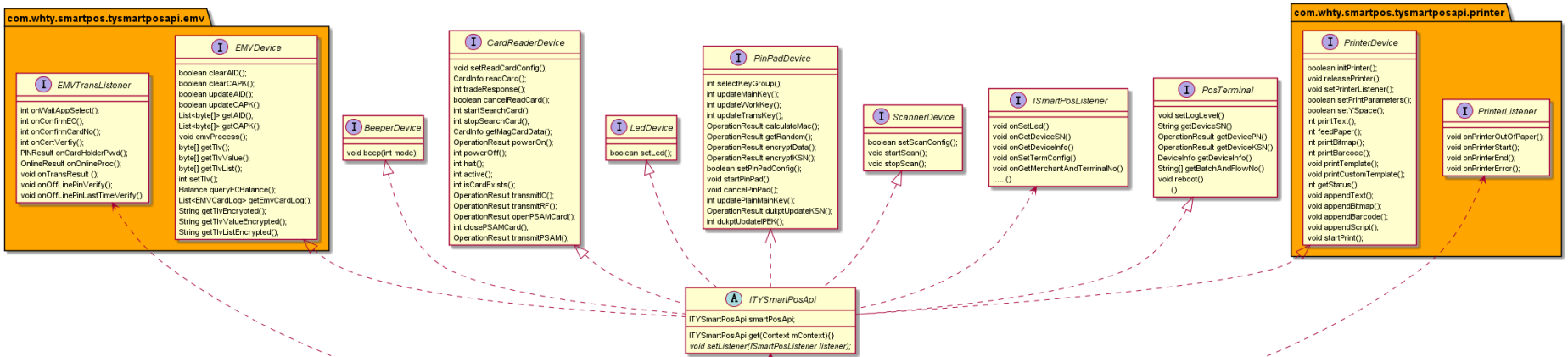
```
}

@Override
public int onCertVerfiy(String certType, String certValue) {
    return 0;
}

@Override
public PINResult onCardHolderPwd(boolean isOnlinePin, int offlinePinLeftTimes) {
    PINResult pinResult = new PINResult();
    pinResult.setResultCode(0);
    pinResult.setPindata(pinblock);
    return pinResult;
}

@Override
public void onTransResult(int code, final String desc) {
   smartPosApi.stopSearchCard();
}
```

```
    @Override
    public void onPbocTerGetInputDisp(byte[] bytes) {
    }

    @Override
    public void onOffLinePinVerify(boolean isPinOK) {
    }

    @Override
    public void onOffLinePinLastTimeVerify(int number) {
    }
}
```

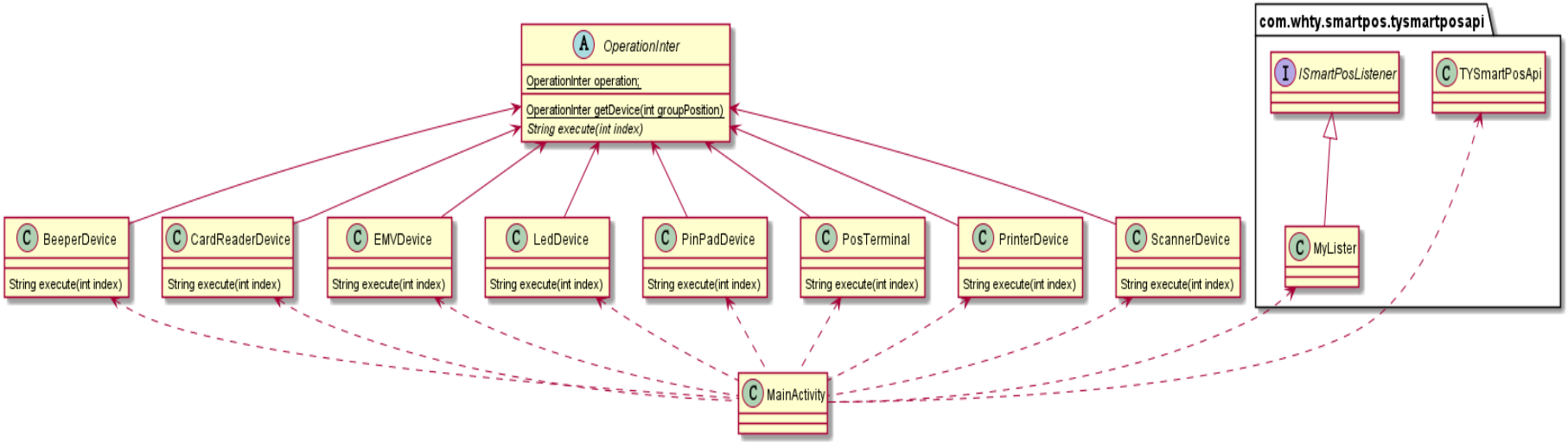
Appendix 3 Function workflow

1. UML diagram

1.1 SDK class diagram

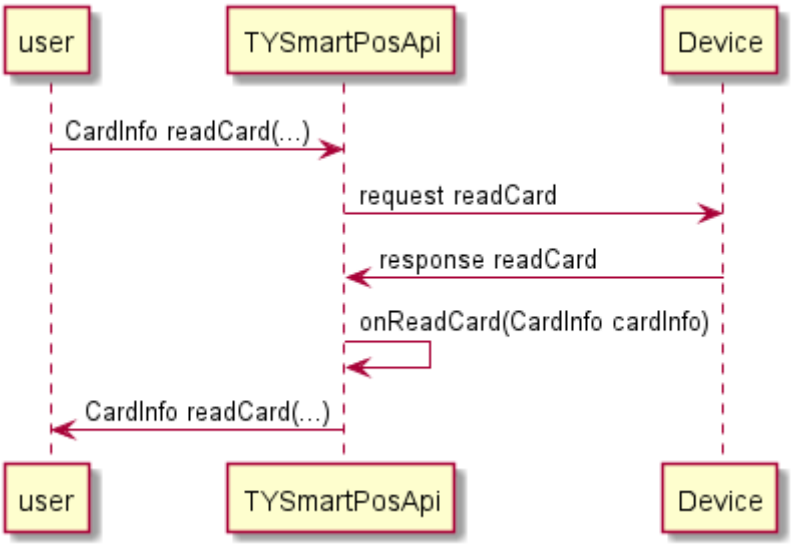


1.2 Demo class diagram

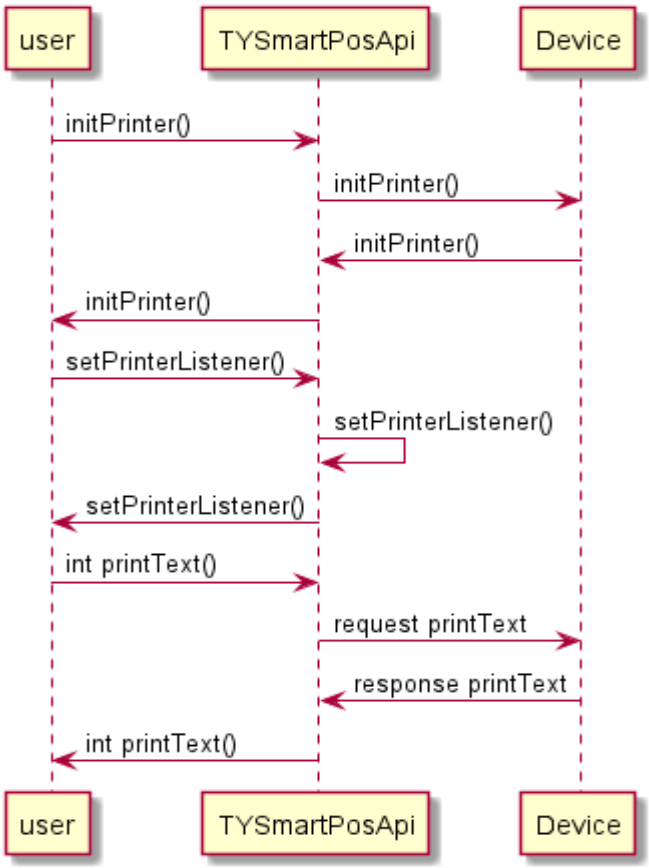

OperationInter
OperationInter operation;
OperationInter getDevice(int groupPosition)
String execute(int index)
com.whty.smartpos.tysmartposapi
ISmartPosListener
TYSmartPosApi
MyLister
BeeperDevice
String execute(int index)
CardReaderDevice
String execute(int index)
EMVDevice
String execute(int index)
LedDevice
String execute(int index)
PinPadDevice
String execute(int index)
PosTerminal
String execute(int index)
PrinterDevice
String execute(int index)
ScannerDevice
String execute(int index)
MainActivity

2. Sequence diagram

2.1 Card reading sequence diagram

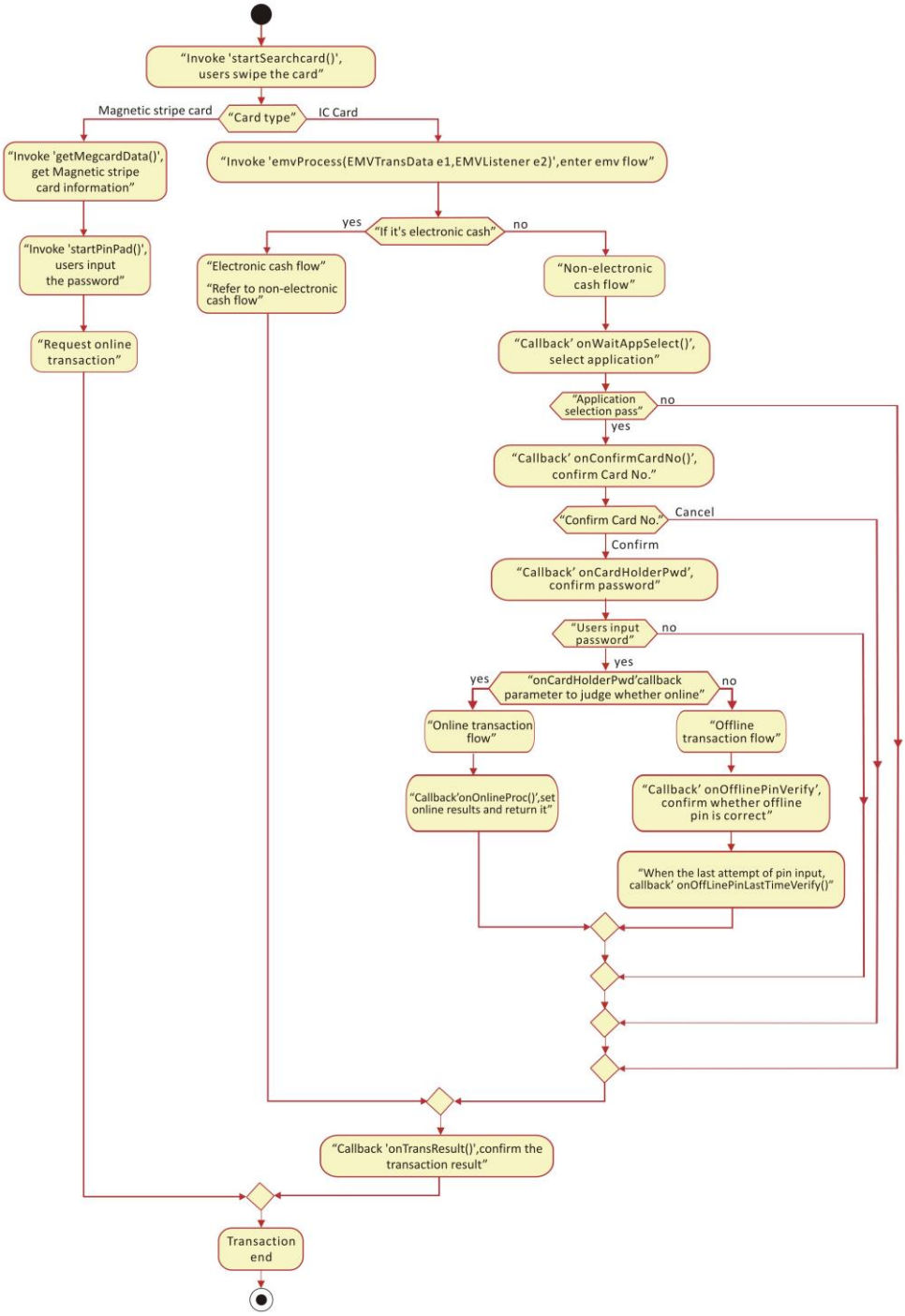


2.2 Printing sequence diagram



3. Flow chart

The transaction flow chart of magnetic stripe card and IC card. After obtaining the card type, users call different interfaces to conduct transactions. The magnetic stripe card obtains the card data through getMagCardData(); the IC card executes the EMV process through emvProcess(). Please refer to the following figure for the standard process (the callback interface of the EMV part may not be triggered due to actual card differences)



Appendix 4 Abbreviations

This chapter explains the abbreviations used in the documentation.

POS	Point Of Sale
SDK	Software Development Kit
SN	Serial Number
KSN	UnionPay Terminal Serial Number/Key Serial Number

APN	Access Point Name
APP	Application
GPS	Global Positioning System
CCID	Chip Card Interface Device
OS	Operating System
MAC	Message Authentication Code
IPEK	Initial Pin Encryption Key
BDK	Base Derivation Key
PSAM	Purchase Secure Access Module
AID	Application Identifier
CAPK	Certificate Authority Public Key
EMV	Europay Mastercard Visa.cooperration
TLV	Tag-Length-Value
EC	Electronic Cash